

Software Requirements Specification

Version 1.0

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Fitness Training Center Management System

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1.0 Introduction	
1.1 purpose	
1.2 Documentation Convention	
1.3 Intended Audience and Reading Suggestion	
1.4 Product Scope	
1.5 Reference	
2.0 Overall Description	
2.1 Product Perspective	
2.2 Product Function	
2.3 User Classes and Characteristics	
2.4 Operating Environment	
2.5 Design and Implementation Constraints	
2.6 User Description	
2.7 Assumptions and Dependencies	
3.0 Extended Inference	
3.1 User Interfaces	
3.2 Hardware Interfaces	
3.4 Communication Interfaces	
4.0 Non-Functional Requirements	
4.1 Compatibility with Other Systems (Accounting and HR)	
4.2 Scalability	
4.3 Secure Users' Personal Data with Stakeholder Authorization	
4.4 Usability	
5.0 Use Case Specification and Use Case Diagram.	
5.1 Use Case Specification.	
5.1.1 User Registration	
5.1.2 Manage Coach Schedule.	
5.1.3 Tracking Coach Performance.	
5.1.4 Managing User Personal Data	
5.1.5 Update Coach Specification.	
5.1.6 Export Financial Reports.	
5.1.7 Update Programs Using AI.	
5.1.8 Schedule Sessions.	
5.1.9 Create Subscription.	
5.1.10 Create Personal Member Programs.	
5.1.11 Manage Waitlist.	
5.1.12 Apply Sessions Schedule Updates.	
5.1.13 Send Subscription Canceling Warning.	
5.1.14 Providing Coach Notes.	
5.1.15 Sending Reminders Notification.	
5.2 Use Case Diagram.	
6.0 Activity Diagrams.	
7.0 Class Diagram.	
8.0 Collaboration Diagrams.	
9.0 Sequence Diagrams.	
10.0 State Diagrams	

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القسم	التقييم	الملاحظات
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1.0 Introduction

1.1 purpose

The purpose of this document is to present a detailed description of the Fitness Training Center Management System. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate. This document is intended for both the stakeholders and the developers.

1.2 Document Convention

Terms	Definition
Coach	A certified professional responsible for designing and delivering nutrition and workout programs to members. Coaches provide guidance, motivation, and feedback to help achieve fitness goals.
Member	An individual who has registered with the fitness center to access its facilities, classes, and services. Members may have different subscription types.
Fitness Training Center	A facility designed for individuals to engage in physical workouts and wellness activities. It may include equipment, classes, and professional coaches to support fitness goals.
Sessions	A single scheduled period during which members participate in a fitness activity, class, or one-on-one training with a coach.
Gymnasium	A space or facility equipped with exercise equipment and designated areas for physical training, group classes, or sports activities
Coach Feedback	A feature that allows coaches to review and track workout programs of a certain member.
Workout Program	Structured schedules designed to help individuals achieve specific fitness objectives. These may include exercises, durations, and progression strategies.
Nutrition Program	Personalized dietary recommendations provided to members to complement their fitness goals. These plans consider dietary preferences, caloric needs, and health conditions.
Workouts	Physical exercises designed to improve fitness levels, of many types including: aerobic, yoga, HIIT(High Intensity Interval Training)
Waitlist	A list of members who want to join a fully booked class or session. Members on the waitlist are notified if spots become available.
Stakeholder	Any individual or group with an interest in the system, including members, coaches, administrators, and external partners (e.g., equipment suppliers or investors).
Center Administrators	An Admin (Administrator) is a user with elevated privileges in a system, responsible for overseeing and managing the operations of the fitness training center. Admins ensure that the system functions smoothly and efficiently by performing tasks

User	An Administrator or a Coach or a Member
Expenses	Costs incurred by the fitness center for operations, such as equipment maintenance, utility bills, staff salaries, and software subscriptions.
Subscription	Membership packages offered to members, defining the access level, duration, and included services. Examples include monthly, yearly, or pay-as-you-go subscriptions.
Permissions	Access rights assigned to users within a system, defining what actions they can perform. Permissions are crucial for maintaining security and operational control
Notifications	Alerts sent to members or coaches regarding session reminders, payment due dates, or system updates.
E-Payments	The digital transfer of funds through online or electronic systems. E-payments are widely used for transactions, allowing members to pay for subscriptions, classes, or services using variety of methods
AI Recommendations	Personalized suggestions for workouts, nutrition, or schedules based on member data and progress analysis.
Coach Specialization	Coach specialization refers to the specific areas of expertise or focus that a fitness coach possesses, such as strength training, yoga, HIIT (High-Intensity Interval Training), rehabilitation, or nutrition planning.
Database	A database is a structured and organized collection of data that allows for efficient storage, retrieval, and management of information.

1.3 Intended Audience and Reading Suggestions:

Developers, project managers, marketing staff, users, testers, and documentation writers

As for the reading suggestions, the next chapter, the Overall Description section, of this document gives an overview of the functionality of the product. It describes the informal requirements and is used to establish a context for the technical requirements specification in the next chapter.

Then, the use cases specification and UML diagrams of this document is written primarily for the developers and describes in technical terms the details of the functionality of the product.

Both sections of the document describe the same software product in its entirety, but are intended for different audiences.

1.4 Product Scope:

This system, designed Fitness Training Center management, connects the Center administration with members and Coaches. It enables members to make payments through various methods, register for sessions, and track their booked sessions and subscriptions through a notification system. Coaches can monitor subscribed members, provide performance-related feedback, and create personalized workout and nutrition programs for each member, which are AI-generated and AI-updatable.

Additionally, the product offers user-friendly administrative tools tailored to center and staff management. These tools allow the administration to monitor expenses and payments (integrated with accounting systems), manage gymnasiums bookings and waitlists, and handle members' personal information and subscriptions. Coaches' details, including specializations, schedules, and performance indicators, can also be managed and accessed (integrated with the HR system).

1.5 References:

IEEE. *IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications*. IEEE Computer Society, 1998.

2.0 Overall Description:

2.1 Product Perspective:

The Fitness Training Center Management System is an integrated solution designed to enhance the operations of fitness centers. It acts as a centralized platform connecting administrators, coaches, and members. The system replaces traditional manual processes with automated features, integrating with accounting and HR systems. The product is self-contained but can connect with external systems like payment gateways and banks.

2.2 Product Functions:

The core functions of the system include:

- Member Management: Registration, subscription management, and profile maintenance
- Session Scheduling: Booking and managing workout sessions, gymnasiums and coaches in charge in addition to support waitlists.
- Personalized Plans: Coach and AI recommended workout and nutrition plans based on member data and updating plan based on coach notes.
- Coach Management: Tracking coach performance, schedules, and assigning members.
- Payment Processing: Supporting e-payments cash and credit cards in addition to generate financial reports.
- Notifications: Alerts for session reminders, subscription renewals, and updates.
- Members Progress Tracking: Evaluation coach notes and tracing member's level

2.3 User Classes and Characteristics:

- Members: Individuals who register to the fitness center and track their workout sessions and programs.
- Coaches: Certified professionals responsible for managing workout and nutrition programs.
- Administrators: Center in charge persons responsible for registering, managing , tracking members and coaches, scheduling session and accessing expenses reports.

2.4 Operating Environment:

- Hardware: Compatible with desktops, tablets, and smartphones.
- Operating System: Windows, macOS, iOS, and Android.
- Web Browser Support: Chrome, Firefox, Safari, and Edge (latest versions).
- Server Requirements: Cloud-based with high availability and fault tolerance.

2.5 Design and Implementation Constraints:

- Specify the limits of administrators, coaches and members' permissions.
- Use standardized frameworks (e.g., React for frontend, Laravel for backend, Flutter for mobile app, MySQL for database).
- Integration with external accounting and HR systems.
- Scalability to support multiple fitness center branches.
- Usability Consideration for users with weak technical expertise.

2.6 User Documentation:

- User Manuals: Guides for members, coaches, and administrators.
- Online Help: Integrated help sections within the web and mobile apps.
- Tutorials: Video guides for key system features.
- Technical Documentation: API references and deployment guides for developers.

2.7 Assumptions and Dependencies:

- Members will have access to smartphones or computers.
- Reliable internet connectivity is assumed for system functionality.
- Dependences on third-party services like payment gateways and cloud hosting providers.
- The HR and accounting system must support API integration for seamless data exchange.
- A recommendation AI system must be coach on fitness training data and ready for integration.

3.0 External Interface Requirements:

3.1 User Interfaces:

- Members: Mobile app for subscription and sessions tracking.
- Coaches: Mobile app with tools for monitoring member progress and updating workout/nutrition plans.
- Administrators: Web dashboard for financial reports, user and coaches management and sessions scheduling.

3.2 Hardware Interfaces:

- Supporting Operating Systems: Windows 7⁺, macOS, and Linux, IOS 16 and Android 6⁺
- Supporting Device Types: PCs with multicore processors, 4GB⁺ RAM and sufficient storage. Smartphones with sufficient processor and ram
- Communication Methods: RESTful APIs over HTTPS

3.3 Software Interfaces:

1. Database Integration: The system uses relational databases (MySQL) to store and retrieve user profiles, session details, and AI-generated plans.

2. Payment Gateway Integration: Secure payment processing services (Stripe) for subscriptions or session bookings.

3. HR System Integration: The system connects with HR platforms (BambooHR) to handle coach data, including schedules, certifications, and metrics.

4. Accounting Tools Integration: Financial management tools (QuickBooks) for exporting financial reports about expenses

5. AI Integration: Recommendations AI system (B-Fit) for providing personalized recommendations for workouts and nutrition plans based on user progress and coach notes.

6. Messaging Services: Cloud messaging Services (Google Firebase) for sending notifications to keep members informed about schedules, bookings, and updates.

3.4 Communications Interfaces:

- Protocols: Secure HTTP (HTTPS) for data transmission.
- Email Notifications: SMTP integration for alerts.
- Messaging Services: Twilio or similar for SMS reminders.
- Data Security: Encrypted communication (e.g., SSL/TLS).

4.0 Non-Functional Requirements:

4.1 Compatibility with Other Systems (Accounting and HR):

The system must integrate seamlessly with external systems such as accounting tools (e.g., QuickBooks, Xero) and HR platforms (e.g., BambooHR, Workday) to streamline workflows, enhance functionality, and reduce manual effort, this requires mainly:

- Data Exchange: Implement RESTful APIs or SOAP protocols to enable real-time data exchange with external systems and standardized data formats (e.g., JSON, XML) for compatibility.
- Accounting Integration: Sync financial data (e.g., payments, subscriptions, refunds) with accounting software, and generate and export detailed financial reports in compliance with local accounting standards.
- HR Integration: Manage coach schedules, payrolls, and certifications through integration with HR systems, and automatically update coach details (e.g., employment status, certifications).
- Error Handling: Handle integration failures gracefully with retry mechanisms and detailed error logging.
- Testing for Compatibility: Conduct end-to-end testing to ensure compatibility with different versions of accounting and HR systems.

4.2 Scalability:

The system must be able to handle increased loads as the fitness center grows, including more branches, members, coaches, and sessions, without significant performance degradation, this requires:

- Database Scalability: Use horizontal scaling techniques (e.g., sharding) to manage growing data volumes for user profiles, sessions, and subscriptions, and support partitioning to handle heavy read/write operations efficiently.
- Application Scalability: Deploy the system on a cloud-based platform (e.g., AWS, Azure, Google Cloud) to leverage elastic resources, and implement load balancers to distribute user requests across multiple servers.
- User Load: Support up to 10,000 concurrent users during peak hours with minimal response time (under 2 seconds).
- Session Management: Manage concurrent session scheduling and notification systems efficiently for multiple branches.
- Monitoring: Use tools like Prometheus or CloudWatch to monitor resource usage and system performance.
- Future Readiness: Ensure modular architecture to allow the addition of new modules or integrations (e.g., IoT fitness devices, analytics tools).

4.3 Secure Users' Personal Data with Stakeholder Authorization:

The system must ensure the security of user data (e.g., personal details, medical history, and payment information) and enforce role-based access control (RBAC) for stakeholders like members, coaches, and administrators.

- Data Protection: Encrypt sensitive data in transit using TLS 1.3 and at rest using AES-256 encryption and comply with data privacy regulations like GDPR or HIPAA (if applicable).
- Authentication: Use secure login mechanisms (e.g., multi-factor authentication) for administrators and coaches.
- Role-Based Access Control (define access levels for each stakeholder):
 - Members: Access their profile, subscription, and workout data only.
 - Coaches: Access assigned members' progress and schedules.
 - Administrators: Full access to system operations and financial data.
- Audit Trails: Maintain detailed logs of all user actions (e.g., data updates, session bookings) for auditing and troubleshooting.
- Threat Detection: Use firewalls, intrusion detection systems (IDS), and regular penetration testing to identify vulnerabilities.
- Backup and Recovery: Implement daily automated backups and ensure recovery processes are tested to restore data within 1 hour of a failure.

4.4 Usability:

The system must be intuitive and user-friendly for all stakeholders (members, coaches, and administrators), minimizing the learning curve and enhancing productivity, this requires:

- User Interface Design: Design a clean and visually appealing UI using standard design principles (e.g., Material Design for web and mobile apps), and use consistent navigation patterns across the system to make it predictable for users.

- **Accessibility:** Ensure compliance with WCAG 2.1 Level AA standards to accommodate users with disabilities, and include features like screen readers, color contrast adjustments, and keyboard navigation.
- **Mobile Usability:** Optimize mobile apps for Android (6.0+) and iOS (16+)
- **Feedback Mechanism:** Provide an in-app feedback option for users to report issues or suggest improvements.
- **Documentation and Tutorials:** Offer detailed user manuals and video tutorials for members, coaches, and administrators, and integrate contextual help within the app (e.g., tooltips, FAQs).
- **Error Messages:** Display user-friendly error messages with clear instructions for resolution (e.g., "Invalid email format" instead of "Error 400").

5.0 Use Case Specification and Use Case Diagram:

Use Case ID	UC-01
Use Case Name	User Registration
Actors	Administrator, HR (for coach registering)
Priority	HIGH
Description	Create a new account for members or coaches by entering their personal data (ID number, name, age, gender, medical history, goals), certificates(coaches).
Preconditions	User doesn't have already an account
Postconditions	User has created his account
Normal Flow	1. Enter data a. For members: personal data (ID number, name, age, gender, medical history, goals) b. For coaches: personal data (ID number, name, age, gender, certificates) 2. Validate data 3. Create account 4. If the user is a coach, send his data to HR. 5. Giving user account's id and password
Alternative Flow	
Exceptions	1- Invalid data 2- User already registered
Assumptions	1. Valid age (15-80) 2. No critical diseases 3. Qualified coaches 4. User will subscribe

Use Case ID	UC-02
Use Case Name	Manage Coach Schedule
Actors	Administrator
Priority	HIGH
Description	Managing coaches working hours, breaks, working days, holydays, logged in time, logged out time also connecting them to sessions schedule
Preconditions	Coach has already registered
Postconditions	A new schedule for the coach is registered to the system
Normal Flow	1. Enter coach available working hours and days 2. Check center schedule 3. Connect coach schedule to sessions schedule 4. Coach Reviews the resulted schedule 5. Coach approves the schedule 6. Registering the final schedule
Alternative Flow	1. Coach refuses the resulted schedule 2. Re-enter coach another available working hours and days 3. Coach rechecks the resulted schedule 4. If the coach working hours didn't fit the center schedule the coach will be registered out of the center
Exceptions	1- No session available 2- Coach's working hours didn't reach the required minimum working hours
Assumptions	1. Coach has time to train 2. The administrator knows the coach specialization

Use Case ID	UC-03
Use Case Name	Tracking Coach Performance
Actors	Administrator, HR, Cloud Messaging Service
Priority	LOW
Description	Tracking coaches by tracking their logged-in time, logged-out time, taken day-offs, commitment to breaks time
Preconditions	1. Administrators have to be logged in. 2. Coach should have at least one active workout program during the current month.
Postconditions	1. A Form sent to HR. 2. Success message to the Administrator. 3. Notification to the coach about.
Normal Flow	1. Enter attended sessions, work days and notes about his performance. 2. Submit the performance form.
Alternative Flow	
Exceptions	
Assumptions	There should be members in the active workout programs for the wanted coach.

Use Case ID	UC-04
Use Case Name	Managing User Personal Data
Actors	Administrator, Accounting
Priority	HIGH
Description	Administrators can view, update, or delete members' and coaches' personal data (ID number, name, age, gender, medical history, goals, certificates).
Preconditions	The user must already exist in the system.
Postconditions	Personal data is updated, deleted, or retrieved successfully.
Normal Flow	1. Retrieve the personal data. 2. Make necessary edits or deletions. 3. Save the changes.
Alternative Flow	If the user doesn't exist, research for the user again.
Exceptions	User doesn't exist.
Assumptions	Personal data is entered correctly.

Use Case ID	UC-05
Use Case Name	Update Coach Specifications
Actors	Administrator, HR, Cloud Messaging Service
Priority	HIGH
Description	Update or redefine coach qualifications, specialties, and certifications.
Preconditions	Coach account exists.
Postconditions	1. Updates will be applied in the start of the next month. 2. Notify the coach and his members.
Normal Flow	1. Modify or add coach qualifications, specialties, and certifications. 2. Save changes 3. Send Changes to HR.
Alternative Flow	Coach must provide another certificate.
Exceptions	1. Invalid or outdated certifications.
Assumptions	1. Certification details are correct. 2. The coach has experience in the updated specialization. 3. has approved on terms and conditions. 4. The coach must wait until the month ends. 5. The coach must update his schedule. 6. Related members must update their subscription data.

Use Case ID	UC-06
Use Case Name	Export Financial Report
Actors	Administrator, Accounting
Priority	MEDIUM
Description	Export Financial report for expenses and profits by entering the date range (from date to date) the admin wishes to see the reports between
Preconditions	the admin must be logged in The accounting department has already entered the operational expenses and subscriptions for given date
Postconditions	
Normal Flow	1. the Admin Enters the date range 2. validate the range 3. fetches the report from the accounting service 4. the system provides the financial reports
Alternative Flow	1. Choose another date 2. revalidate 3. re-request the new report
Exceptions	The range date is invalid The requested report has been not fully prepared yet
Assumptions	1. the user must be admin 2. the admin must be authorized to see financial reports

Use Case ID	UC-07
Use Case Name	Update programs using AI
Actors	AI, Coaches
Priority	HIGH
Description	The system uses AI to automatically update workout and nutrition programs for members based on their progress, performance data, and feedback from coaches. This ensures personalized and adaptive plans for members' fitness goals.
Preconditions	Member profile exists in the system with complete fitness data (e.g., goals, medical history, workout history). Sufficient progress or performance data is available for analysis. AI models for generating recommendations are active and trained. new coach notes (optional)
Postconditions	1. The updated program is saved in the system. 2. Members are notified of the updated plan. 3. Progress data is archived for future updates.
Normal Flow	1. coach add or update the member progress and give notes on the member. 2. The AI retrieves the member's progress, coach notes and past workout/nutrition data. 3. It analyzes the progress data, including completed sessions and performance metrics. 4. The AI generates updated recommendations for the workout and nutrition plans. 5. Coach validates the AI-generated recommendations with predefined fitness rules. 6. The updated program is saved to the member's account. 7. Notifications are sent to the members about the updated program.
Alternative Flow	1.1-If insufficient data exists for analysis: 1.2-The system requests additional input from the coach (e.g., recent activities or updated goals). 2- After receiving the required data, the AI reprocesses the update. 3.1if the coach rejected the updated program 3.2AI will regenerate another program.
Exceptions	1. Invalid or incomplete progress data that cannot be processed. 2. coach rejected the updated program.
Assumptions	1. coach regularly logs the member's progress data, such as completed workouts or dietary information. 2. Coaches timely give notes on member progress to improve AI recommendations. 3. The AI models are up-to-date, well-trained, and capable of handling diverse fitness requirements. 4. coach notes should be suitable for member.

Use Case ID	UC-08
Use Case Name	Schedule sessions
Actors	Administrator
Priority	HIGH
Description	Scheduling sessions by specifying days, time of day, gymnasium, by adding new ones, update or deleting old ones
Preconditions	the admin must be logged in
Postconditions	1. The session must be saved to the system DB 2. The center schedule will be updated at the start of the next month
Normal Flow	1- System suggests available sessions to be scheduled 2- The admin chooses desired sessions from suggested ones 3- The sessions have been scheduled successfully
Alternative Flow	The admin should update or delete sessions ones to add new ones
Exceptions	No available sessions to be scheduled at all
Assumptions	The scheduled session should be suitable for the selected workout type by gymnasium

Use Case ID	UC-09
Use Case Name	Create subscription
Actors	Administrator, Bank, Cloud Messaging Service
Priority	HIGH
Description	Subscribe to a member under a workout program in specific sessions, gymnasium and the coach.
Preconditions	1. Members should already have an account in the system. 2. Administrators should be logged in.
Postconditions	1. Linking members with the workout program with sessions, gymnasium and coach in the coach in system DB. 2. Send notification to members and coach. 3. Members will be subscribed in the workout program 4. The subscription cost will be withdrawn from the member budget.
Normal Flow	1. Suggest to member workout program. 2. Choose the workout program. 3. Suggest sessions and coaches. 4. If the member chooses a specific coach, sessions will be suggested and vice versa. 5. Assign available gymnasium. 6. Ask members to verify the subscription result. 7. member approved. 8. Ask members to pay by available payment methods (e-wallet, credit card, cash). 9. Verify member sufficient money. 10. Give the member the subscription card, and the schedule. 11. Provide the bill.
Alternative Flow	1.1 Member charges his e-wallet and try again. 1.2 pay cash instead. 2.1 Change workout program. 2.2 Add members to waitlist without paying by taking member name and phone number. 2.3 Cancel the process.
Exceptions	Not enough balance (when e-pay or credit-card method): Subscription is not available by the member chosen specifications
Assumptions	Valid age, health, gender for the workout program. 1. Members should have charged credit cards or charged e-wallet or sufficient cash. 2. No diseases prevent members from working out. 3. Members will attend the workout program. 4. E-bay systems are working effectively.

Use Case ID	UC-10
Use Case Name	Create Personal member Programs
Actors	Coach, AI, Cloud Messaging Service
Priority	HIGH
Description	Generate personalized workout and nutrition program for members by AI.
Preconditions	Member profile newly created.
Postconditions	Recommendations are generated and displayed. Notification with new recommended program for member will be sent. Save recommended program to a member profile.
Normal Flow	1. AI Retrieves the member's profile and fitness data. 2. AI Analyzes goals and progress. 3. AI generates a workout and nutrition program. 4. Coach Approve the suggested plan.
Alternative Flow	AI Reanalyze and make a new program coach reapprove the recommended program.
Exceptions	Coach rejected recommended program.
Assumptions	Member data is up to date.

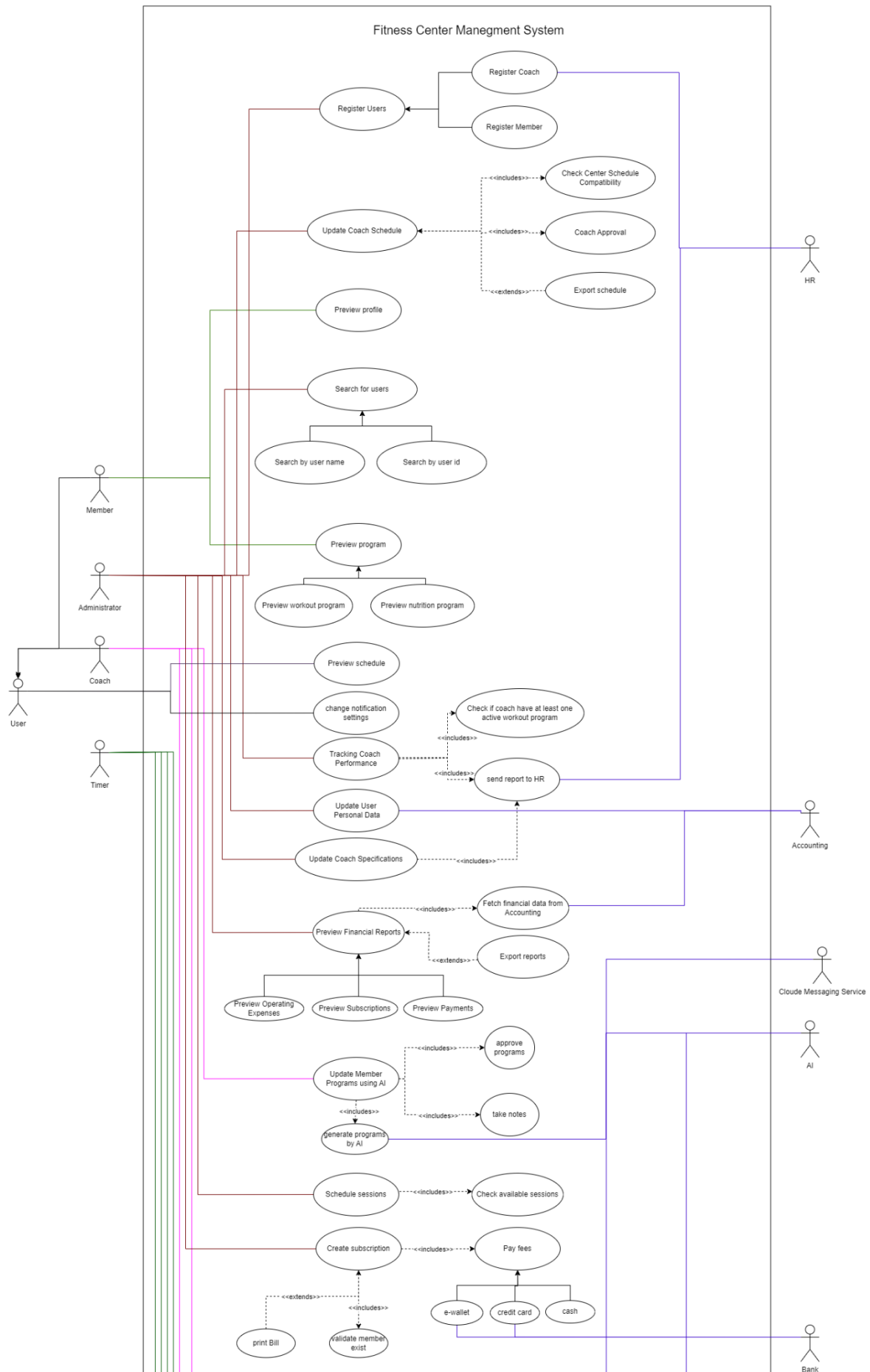
Use Case ID	UC-11
Use Case Name	Manage Waitlist
Actors	Timer, Cloud Messaging Service
Priority	MEDIUM
Description	The action of getting members from the waitlist queue and subscribe them in the wanted workout program.
Preconditions	There should be members on the waitlist.
Postconditions	Waitlist size will be updated in system DB.
Normal Flow	1. Get the first enrolled member in the waitlist queue. 2. Check if there is a place in his workout program. 3. Send notification to members about his subscription availability. 4. Get the next member and repeat the process.
Alternative Flow	Keep the member on the waitlist and discuss the next member.
Exceptions	Subscription is still not available by the chosen specification.
Assumptions	1. Valid age, and health for the workout program when popping up members. 2. No diseases prevent members from working out. 3. Members will attend the workout program.

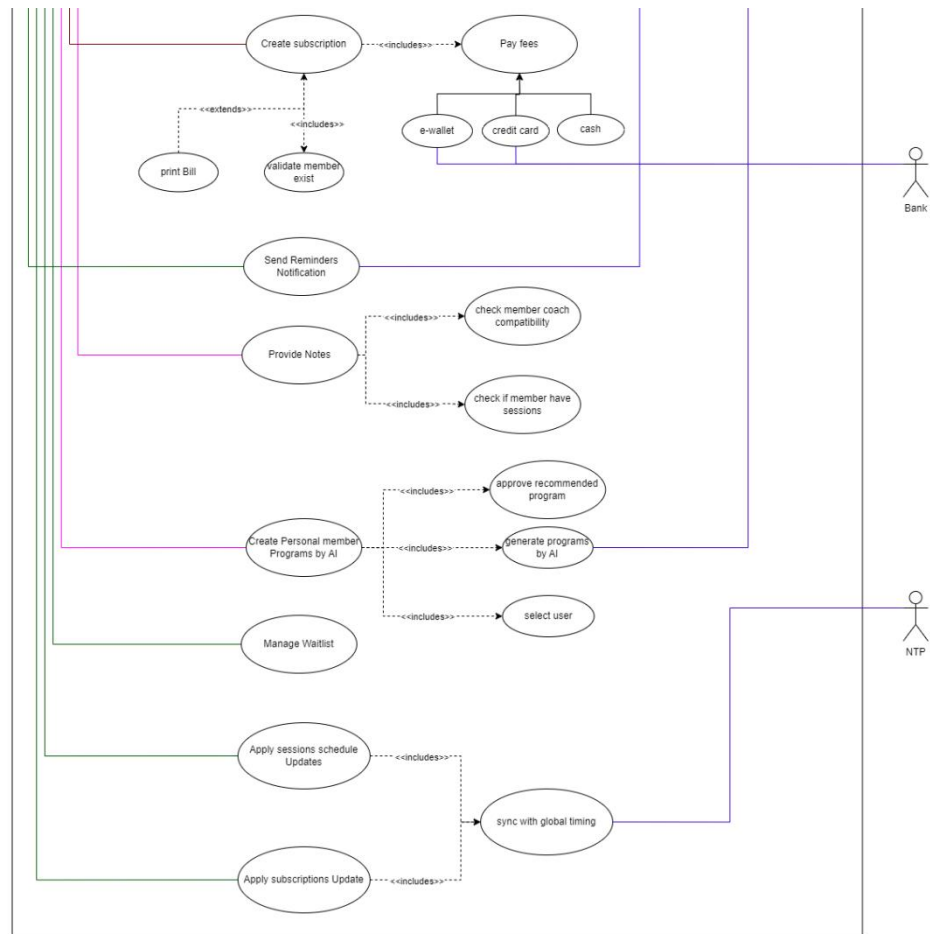
Use Case ID	UC-12
Use Case Name	Apply sessions schedule Updates
Actor	Timer, Cloud Messaging Service, NTP
Priority	HIGH
Description	update center's schedule
Preconditions	Server should sync with NTP Servers
Postconditions	Write about changes to system DB
Normal Flow	1. Apply new sessions schedule updates to center schedule 2. Send notifications to admins and coach about the updated schedule
Alternative Flow	
Exceptions	No updates to be applies
Assumptions	Sessions schedule already updated by admin

Use Case ID	UC-13
Use Case Name	Send subscriptions canceling warning
Actor	Timer, NTP, Cloud Messaging Service
Priority	HIGH
Description	notify members about their canceled subscriptions
Preconditions	Server should sync with NTP Servers
Postconditions	
Normal Flow	1- Fetch Center members. 2-check if their subscription renewed. 3- Send notification to each member from them
Alternative Flow	
Exceptions	Some users haven't renewed their subscription.
Assumptions	

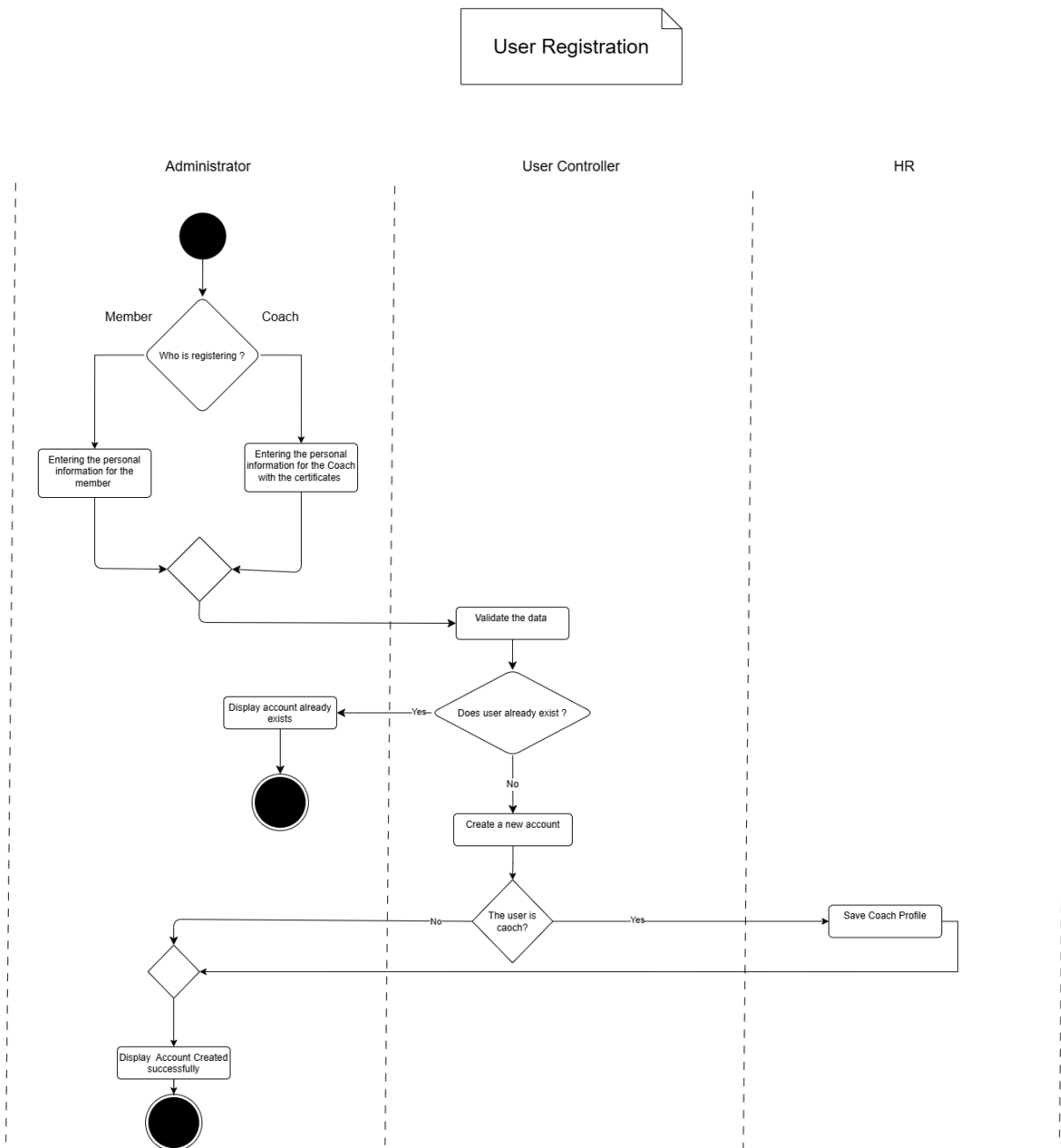
Use Case ID	UC-14
Use Case Name	providing coach notes
Actors	Coach
Priority	MEDIUM
Description	Providing notes about workout status of the member and nutrition status as well
Preconditions	1. The member exists in the system 2. The member training under the supervision of this coach 3. The coach must be logged in
Postconditions	1. The notes must be saved to the system DB 2. The Recommendation AI must be triggered to generate new programs
Normal Flow	1. the coach enters the notes 2. save the notes
Alternative Flow	research with another name or id
Exceptions	The searched member not found
Assumptions	1. The members must attend at least one session 2. The member must be attending sessions at the time

Use Case ID	UC-15
Use Case Name	Sending reminders notification
Actors	Cloud Messaging Service, Timer
Priority	HIGH
Description	Automatically send notifications to members or coaches for reminders.
Preconditions	Notifications must be configured in the system.
Postconditions	Notifications are delivered successfully.
Normal Flow	1. Check if the time hit predefined stored time. 2. Fetch users who already activated notification service. 3. Generate the notification content. 4. Send the notification.
Alternative Flow	
Exceptions	No users activated their notification service.
Assumptions	Sent notification's content are correct.

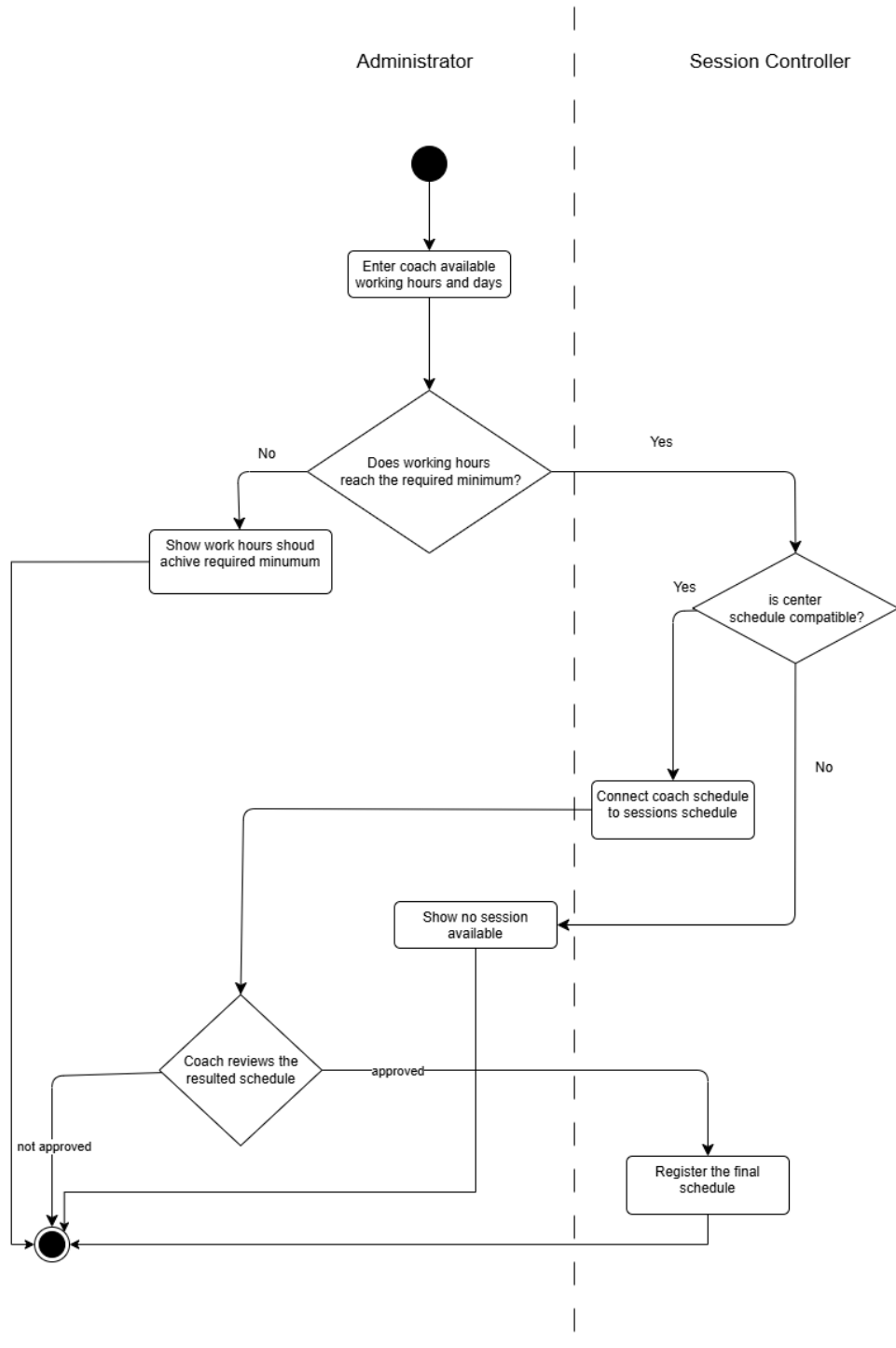




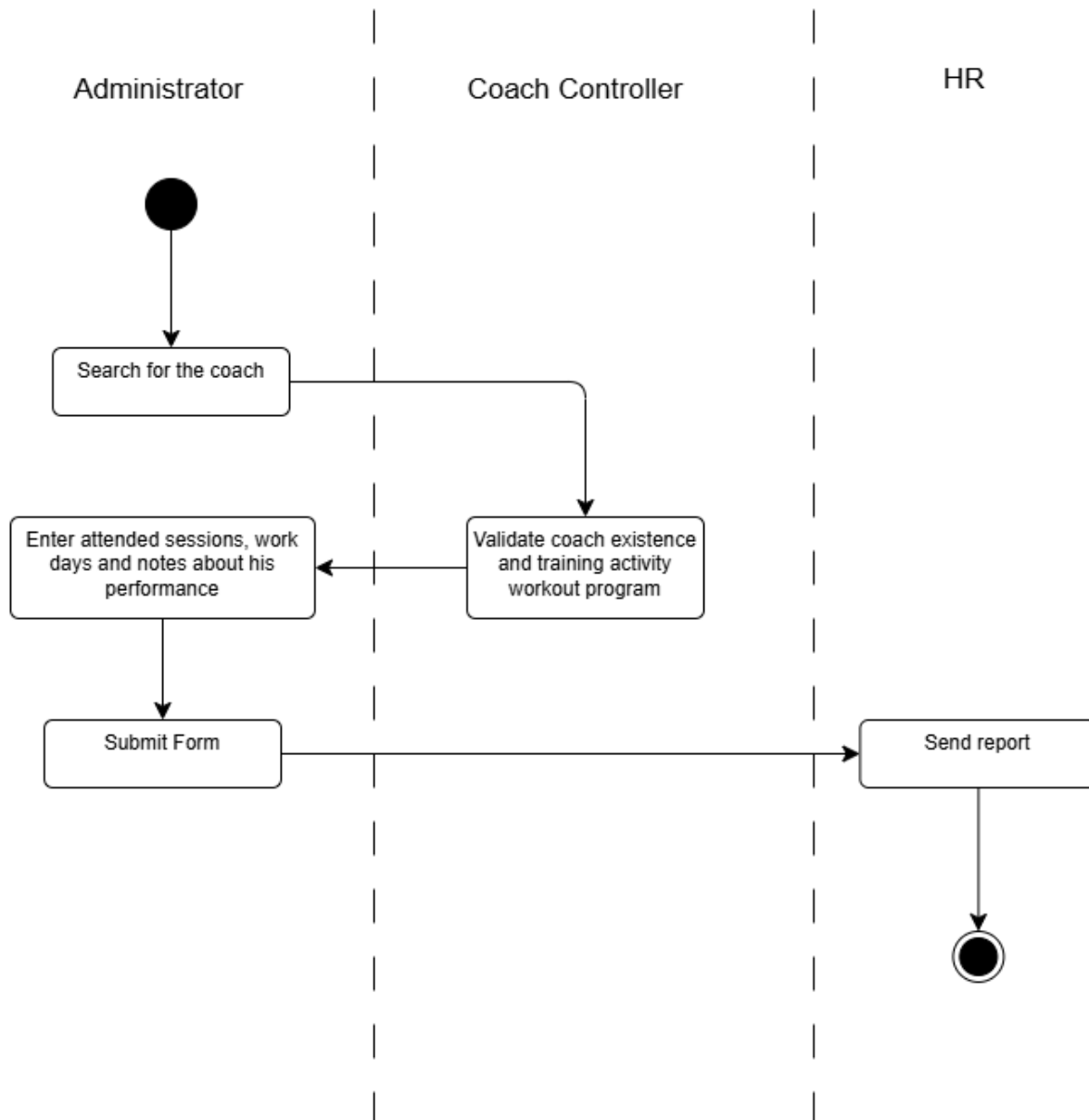
6.0 Activity Diagrams:



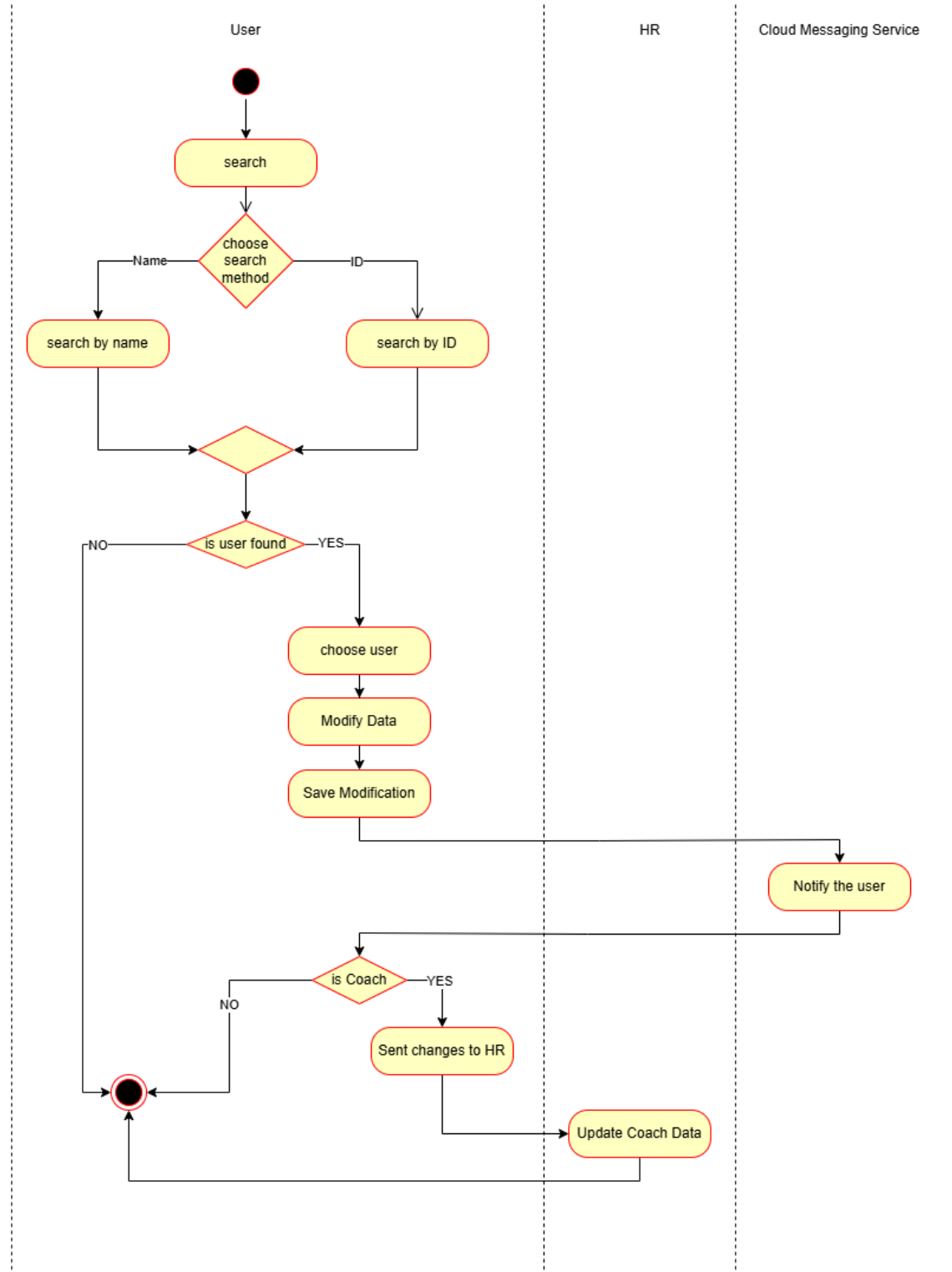
Manage Coach Schedule

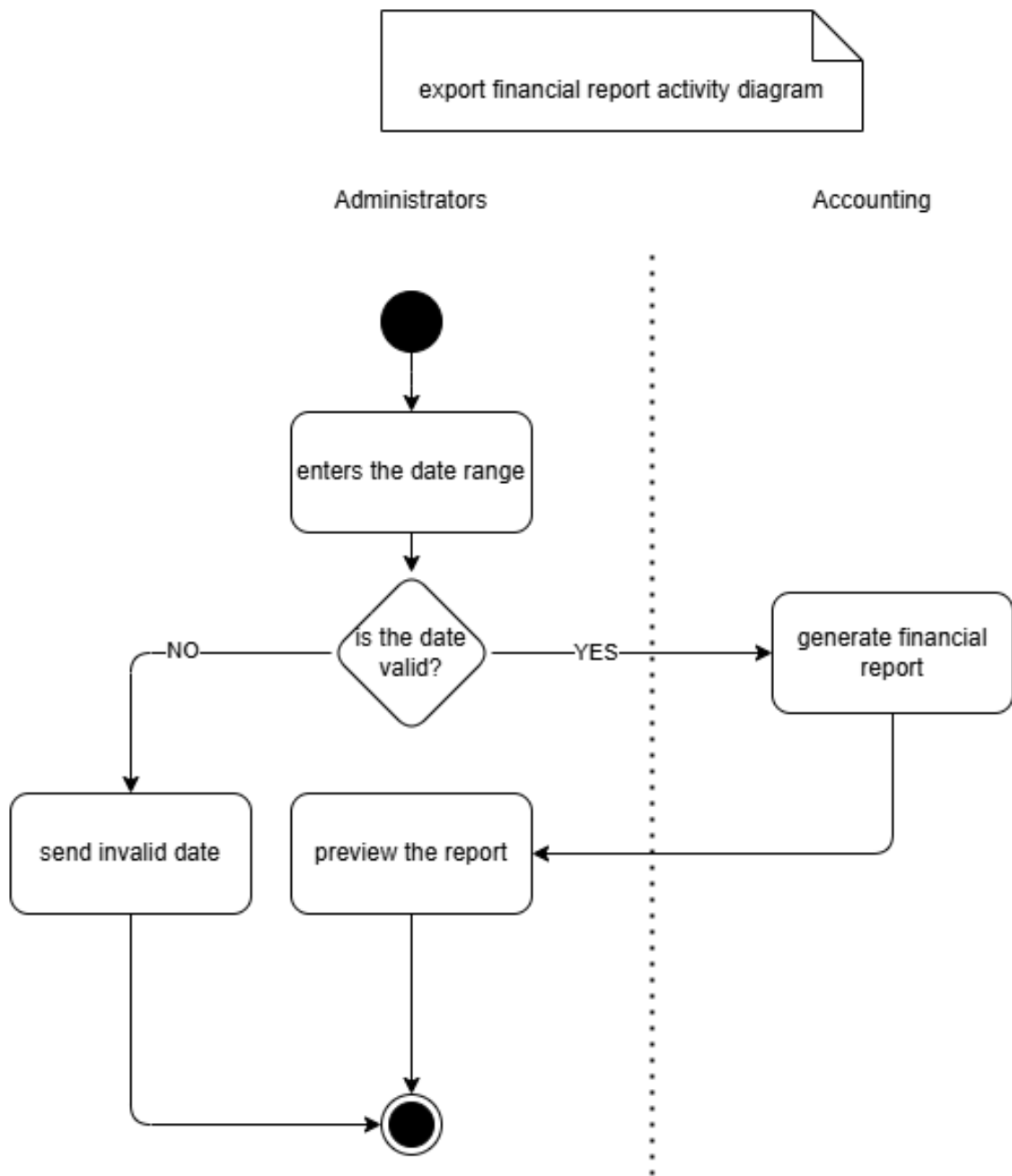


Tracking Coach Performance

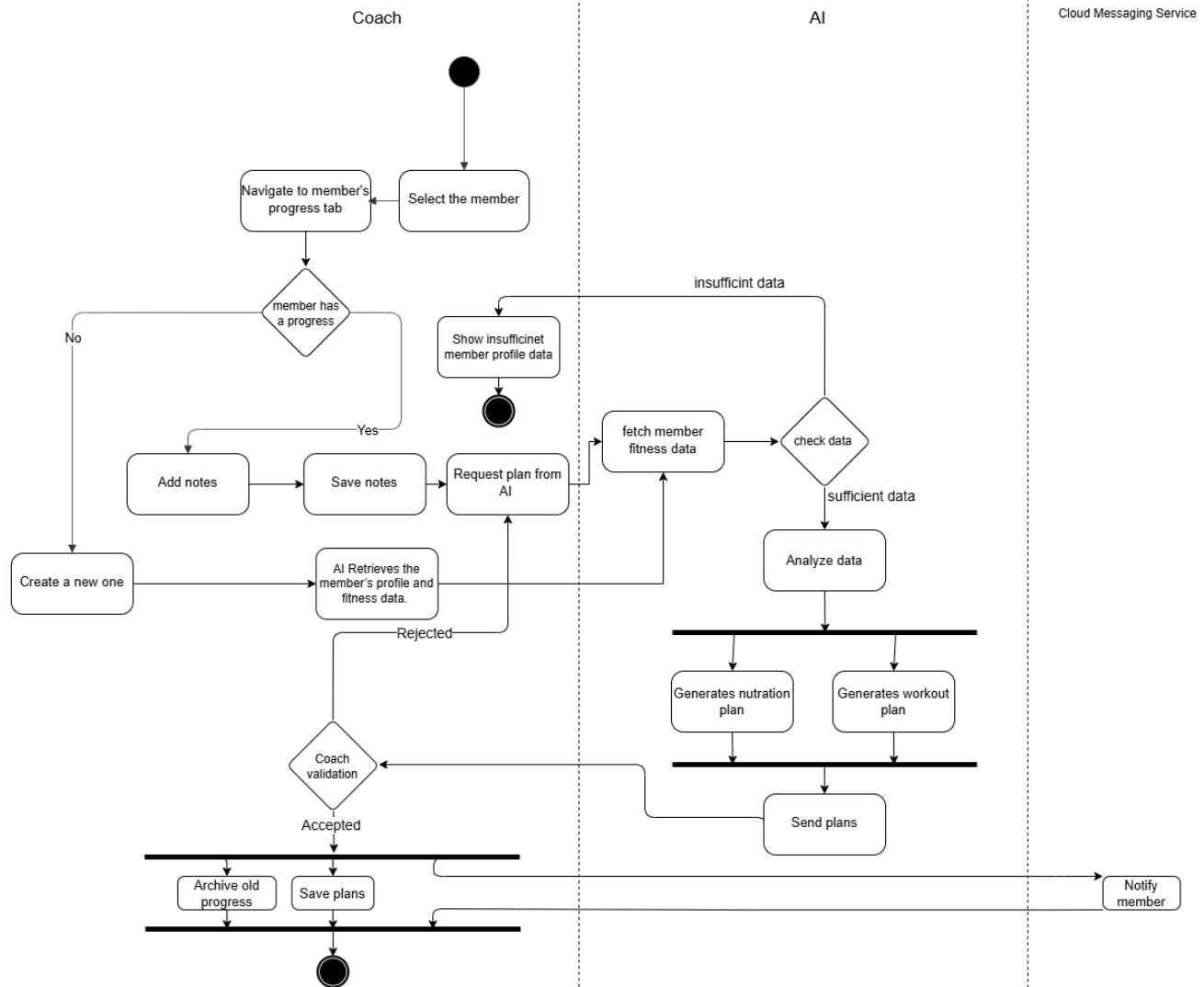


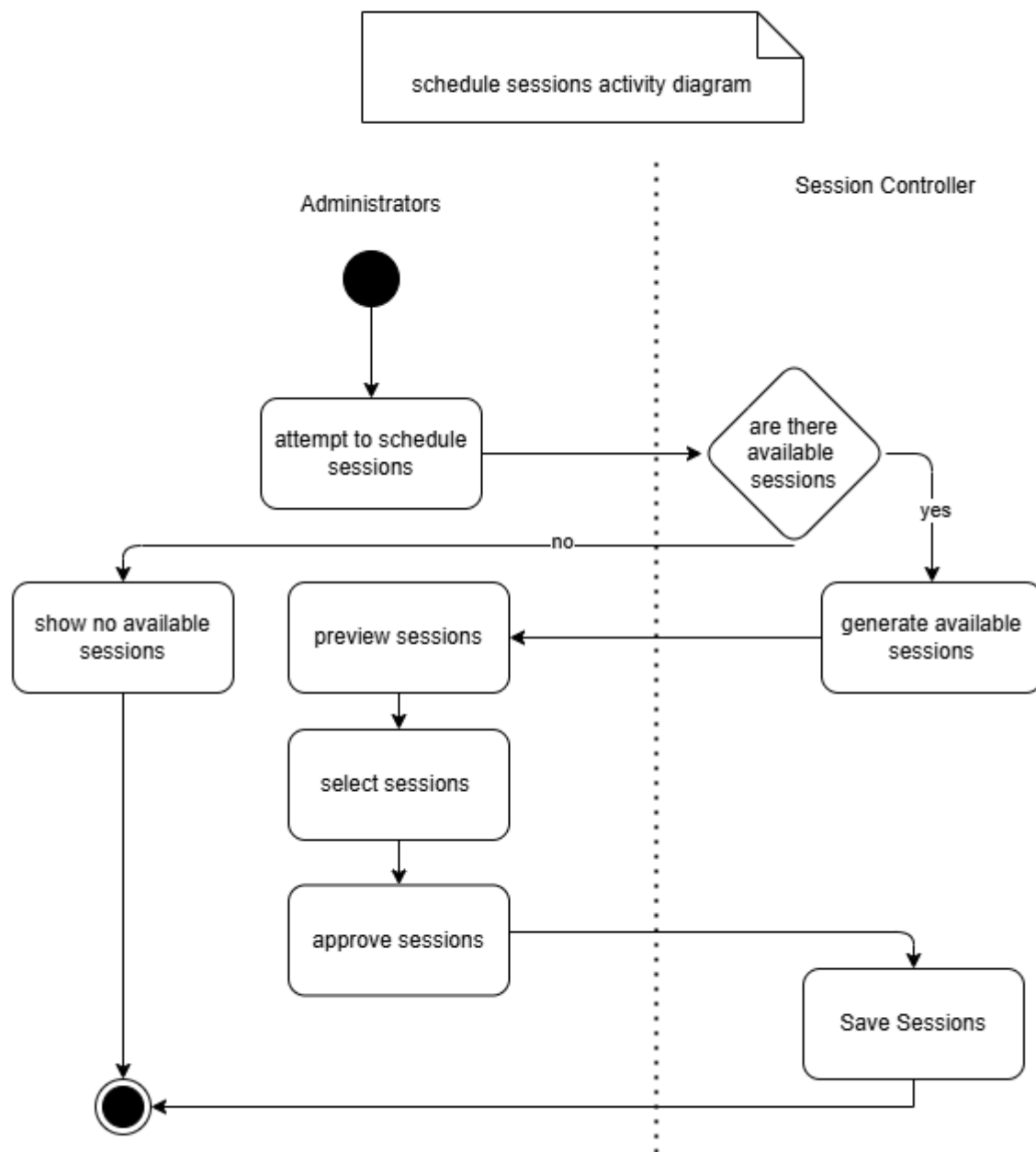
Managing User Personal Data & Update Coach Specifications



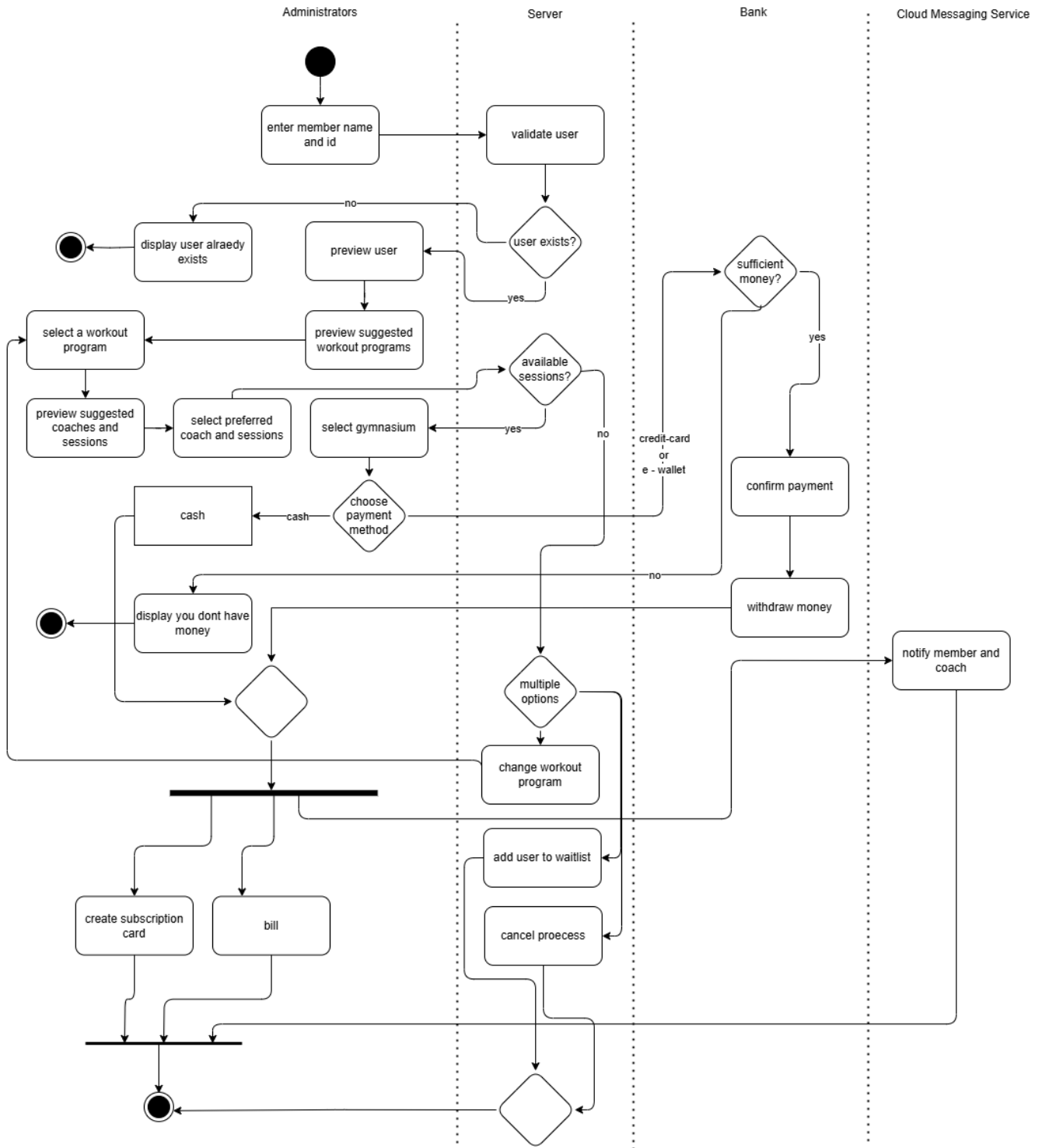


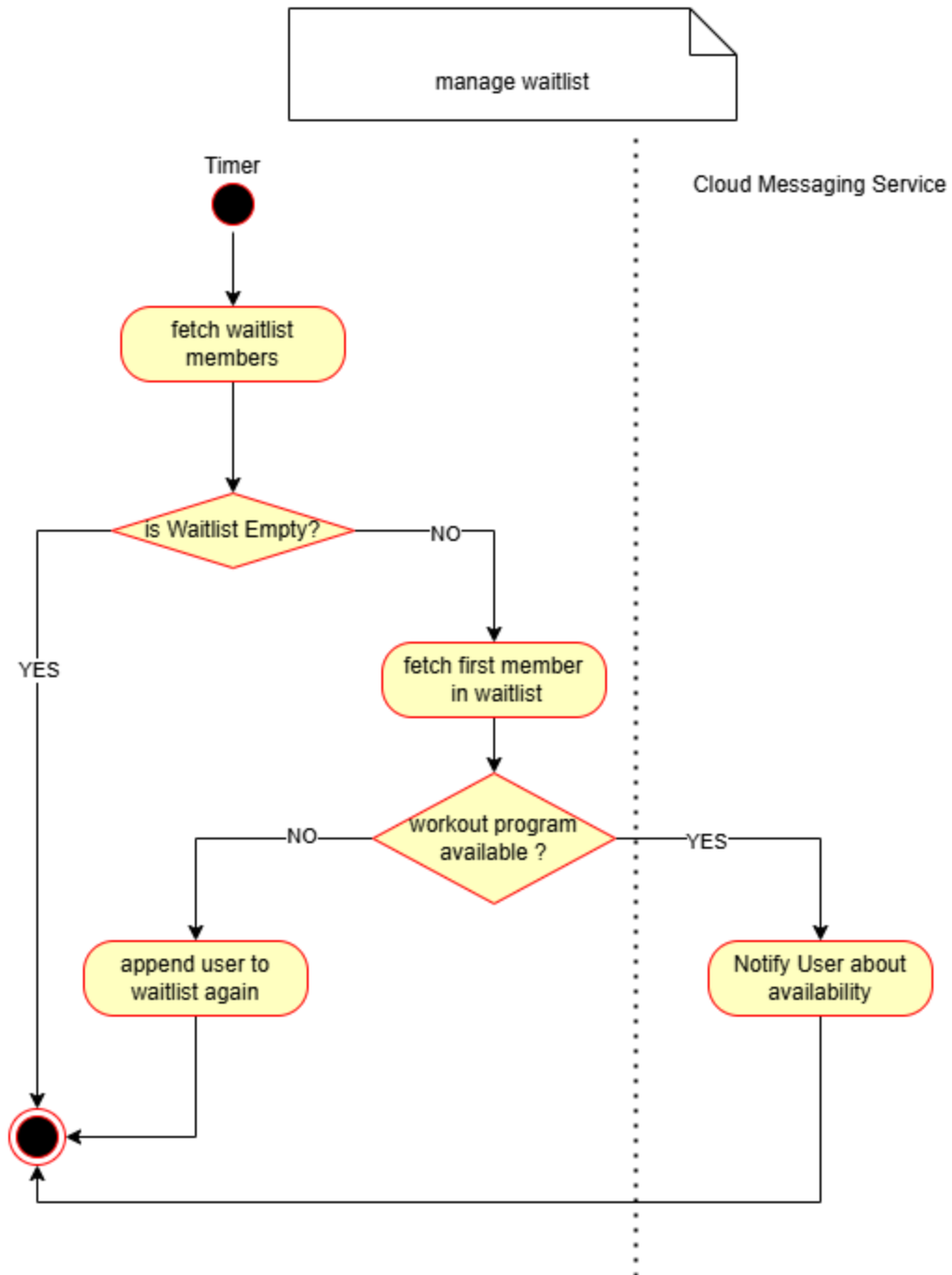
Create and Update programs by AI



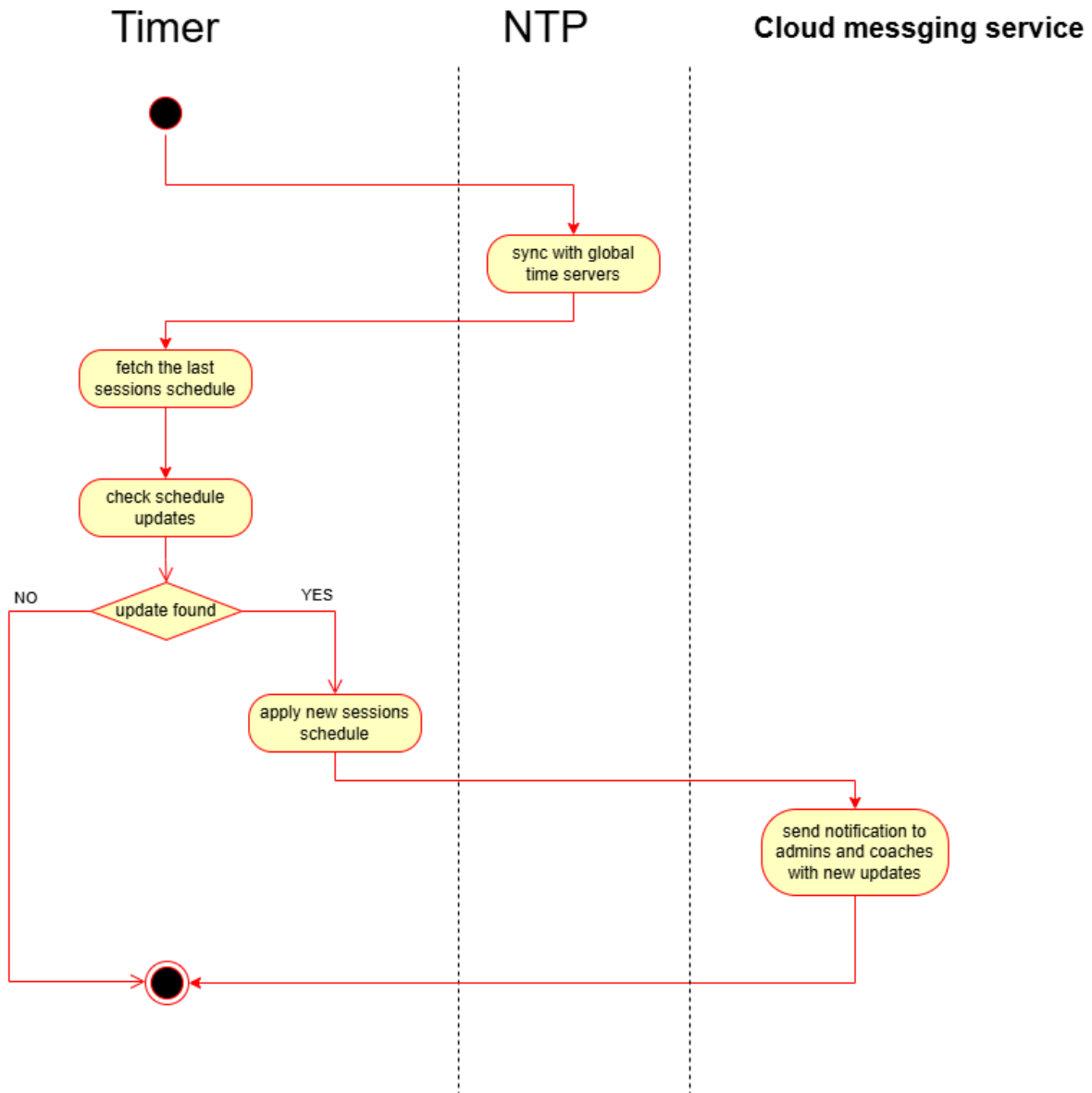


create subscription activity diagram

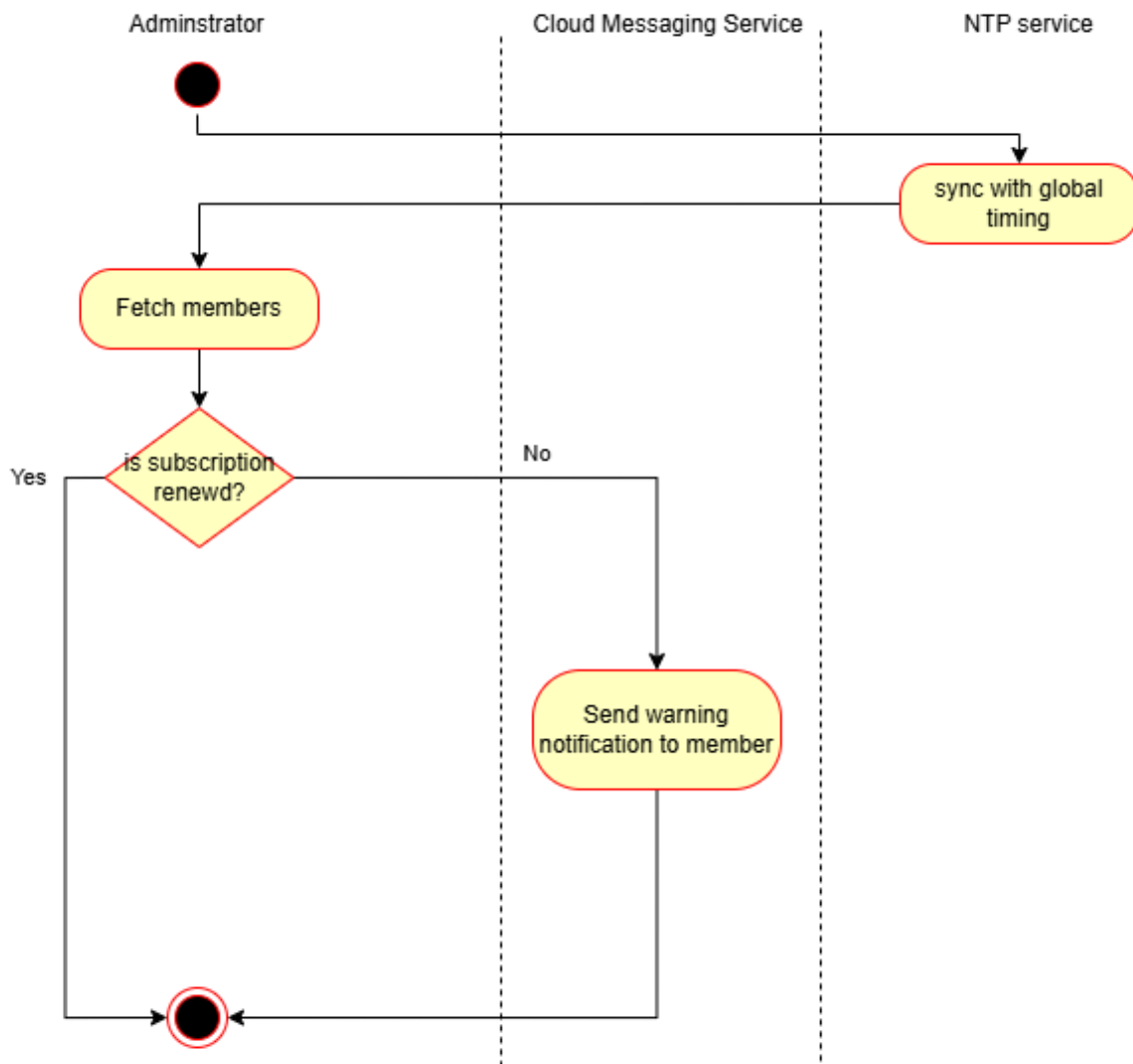




Apply sessions schedule Updates

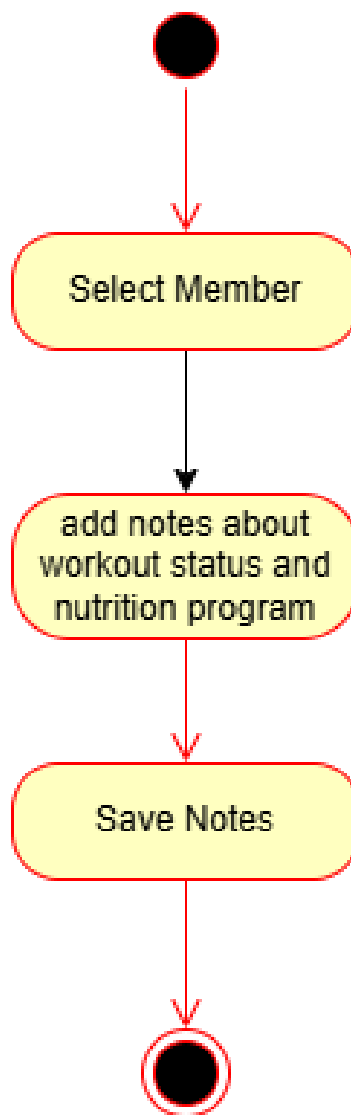


Send subscriptions canceling warning

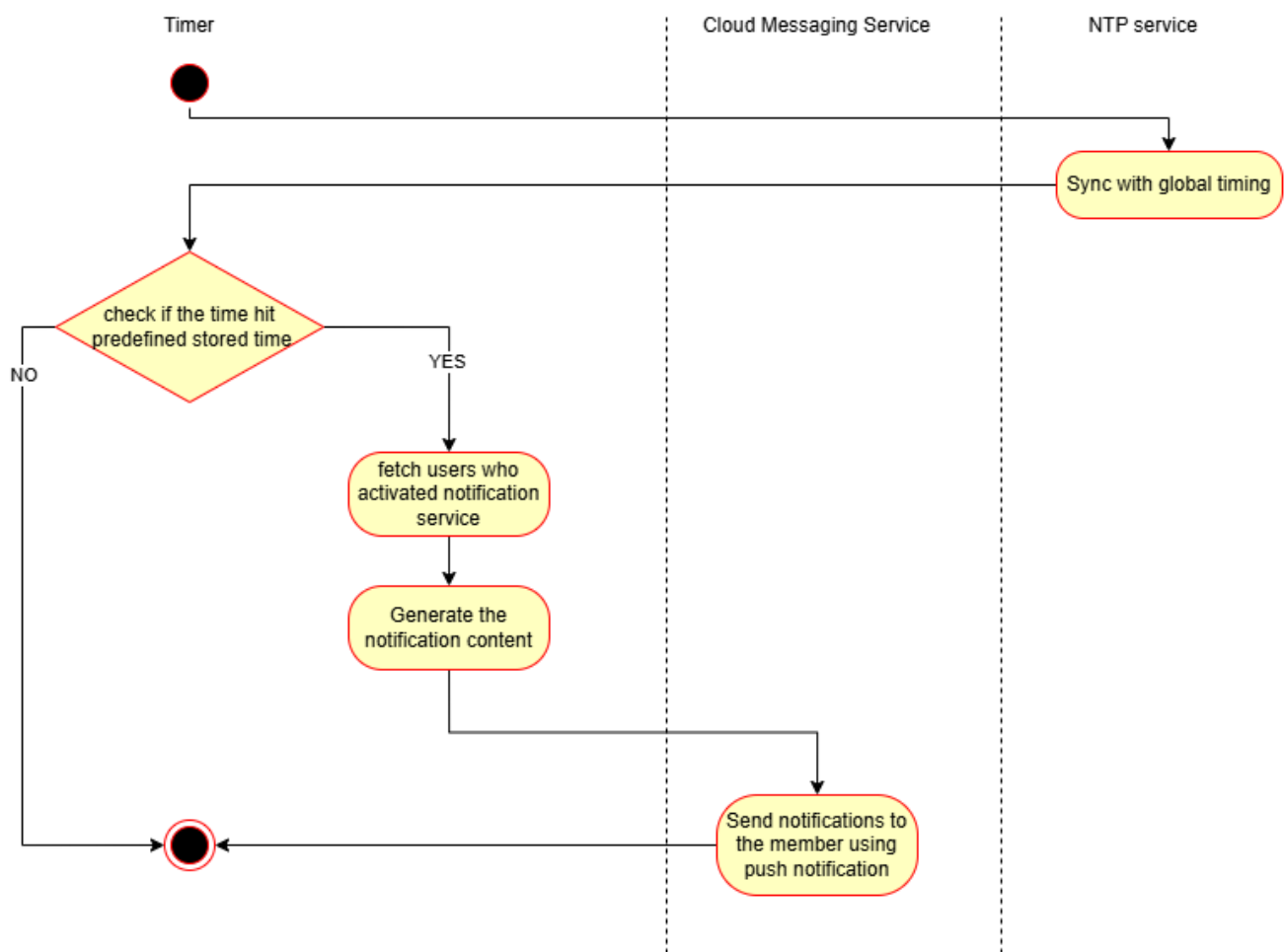




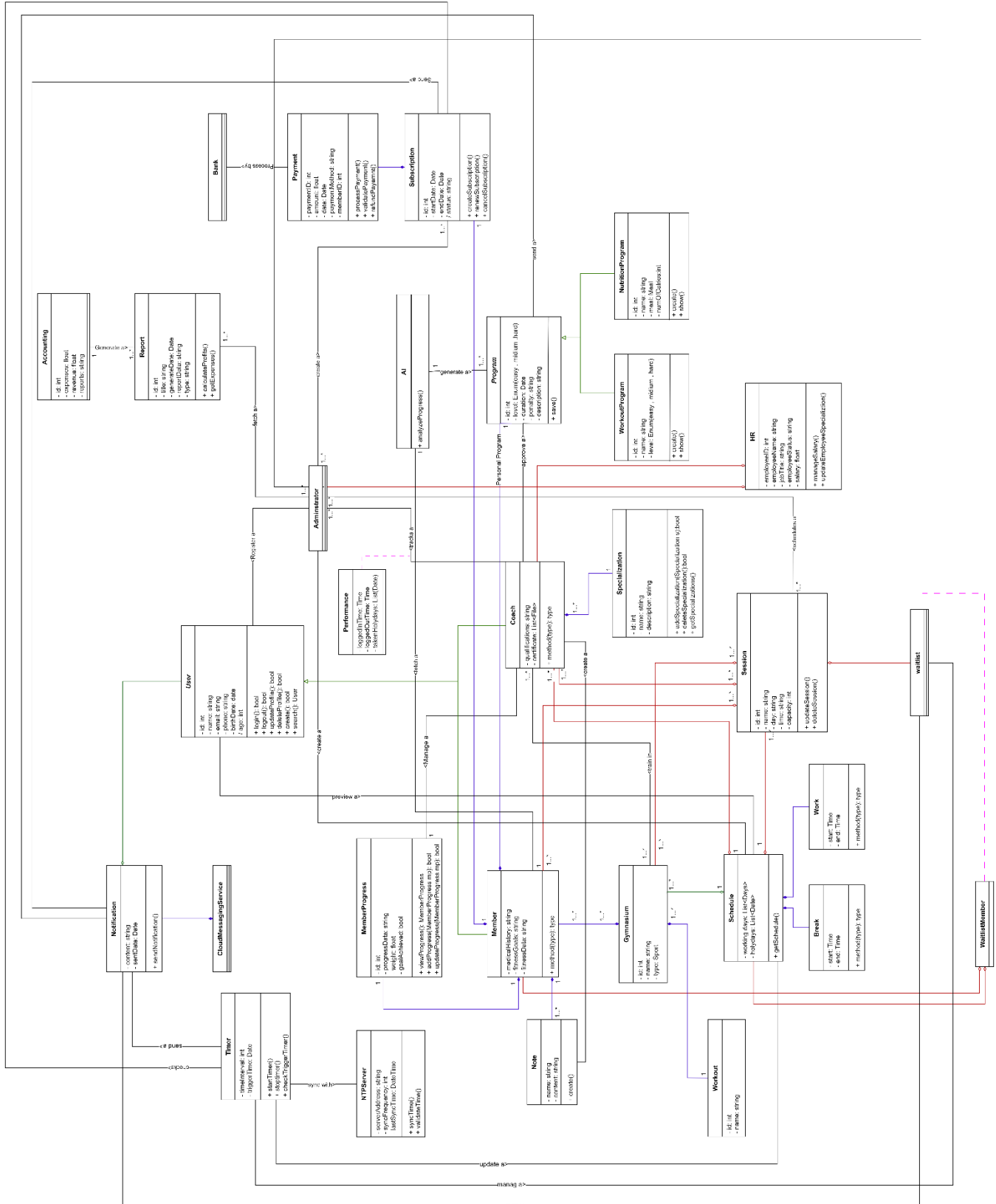
Coach



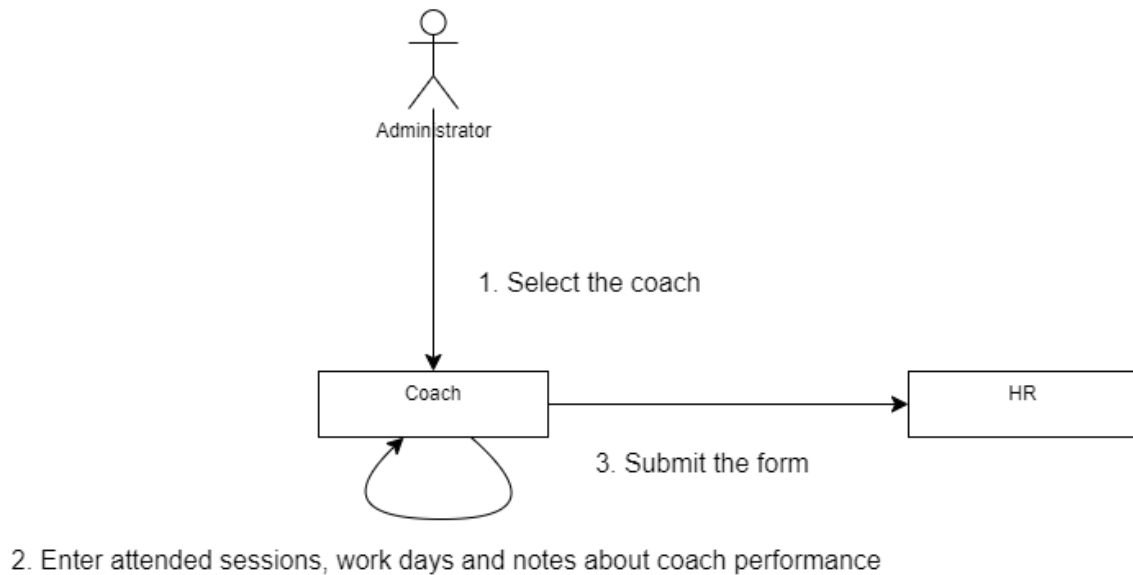
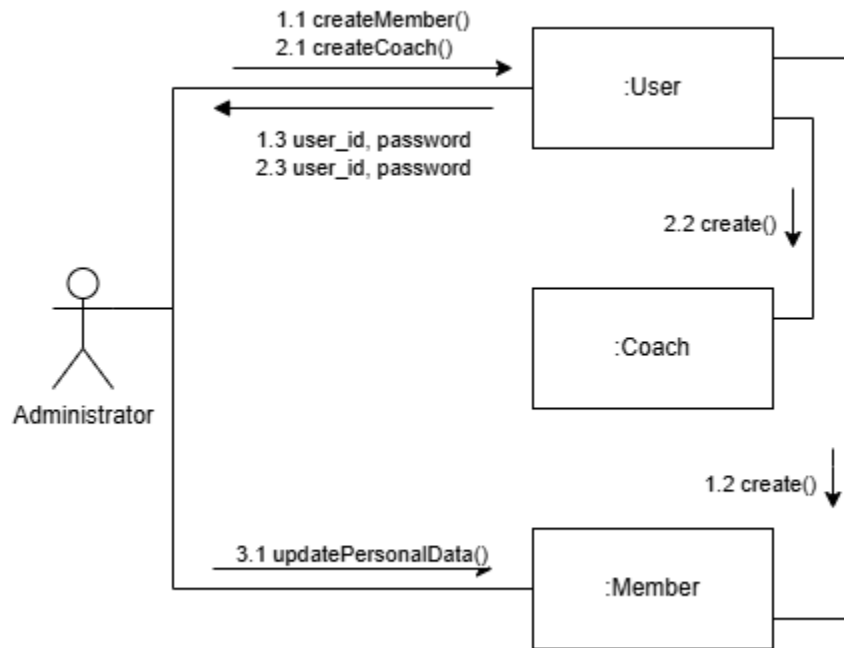
Sending Reminder Notifications

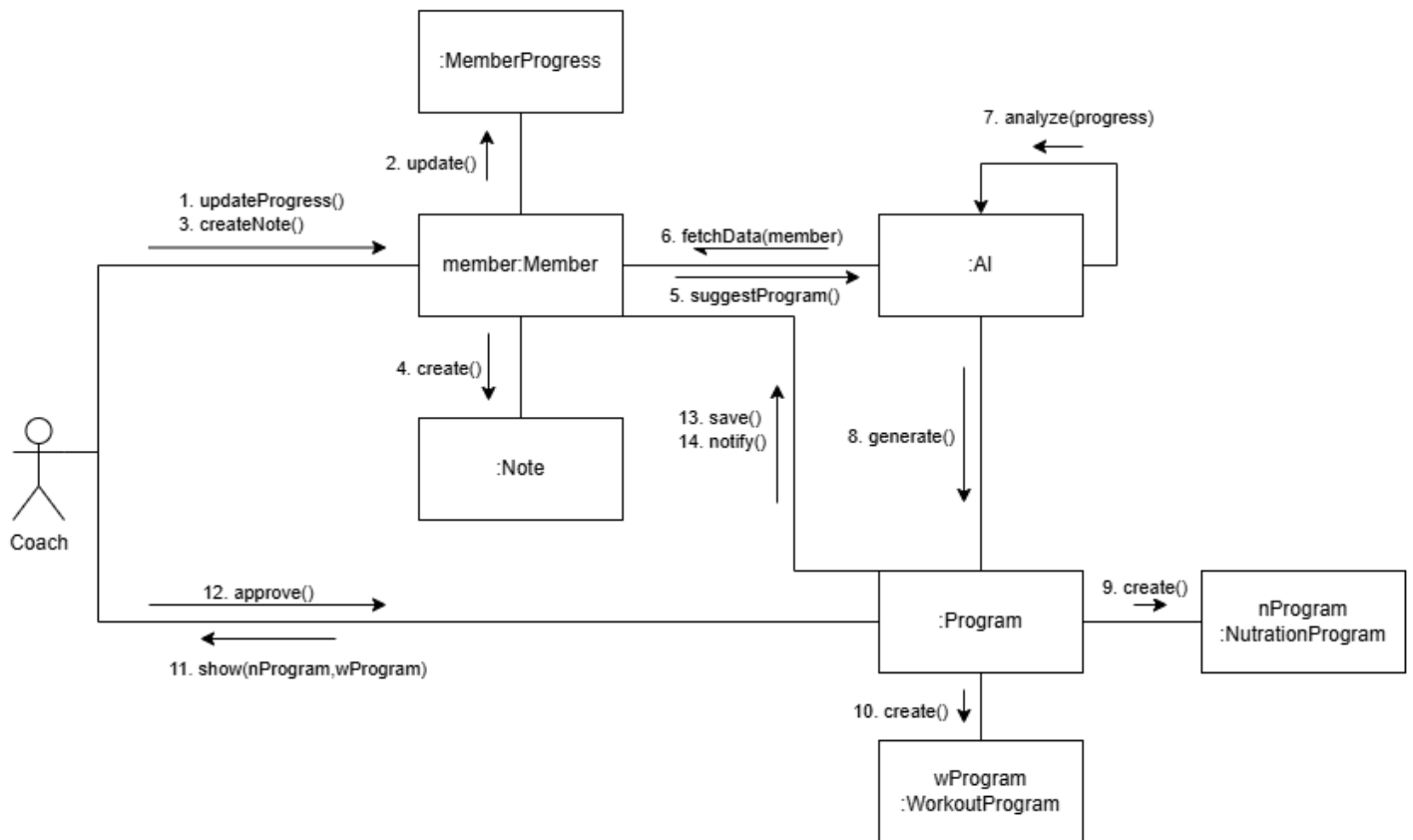
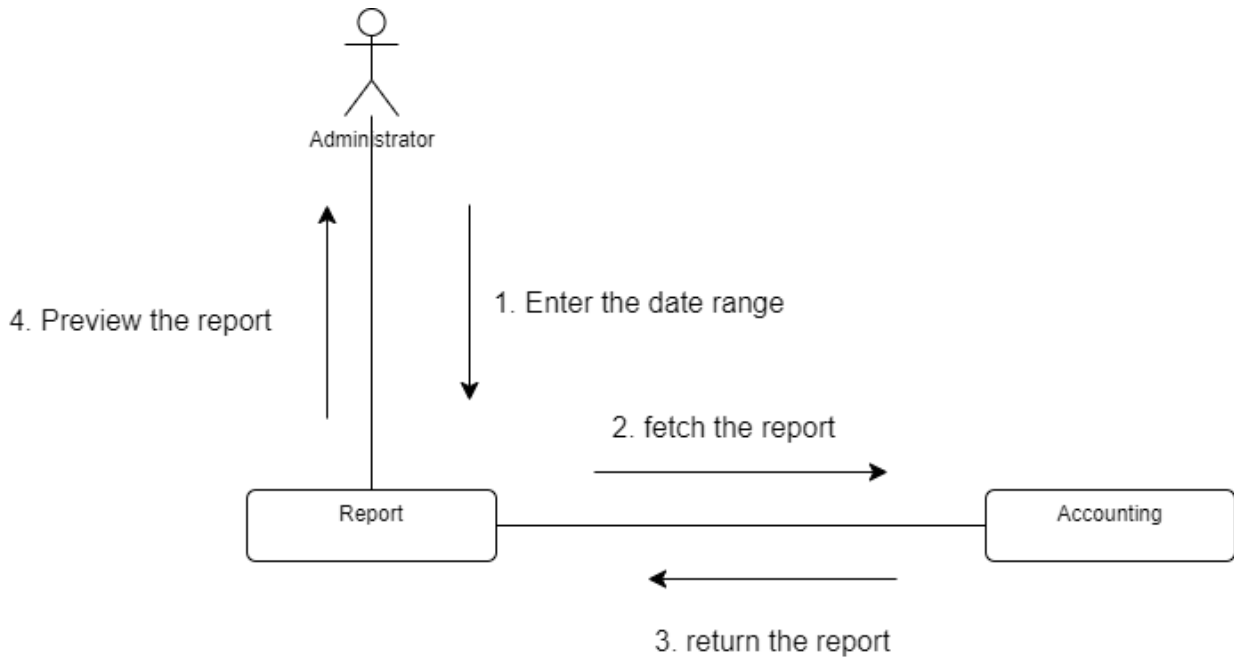


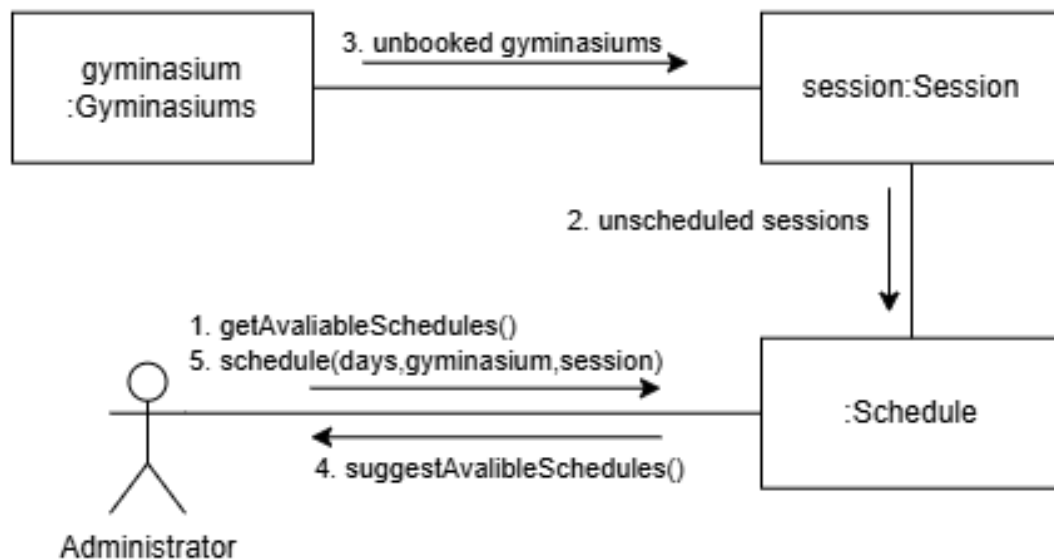
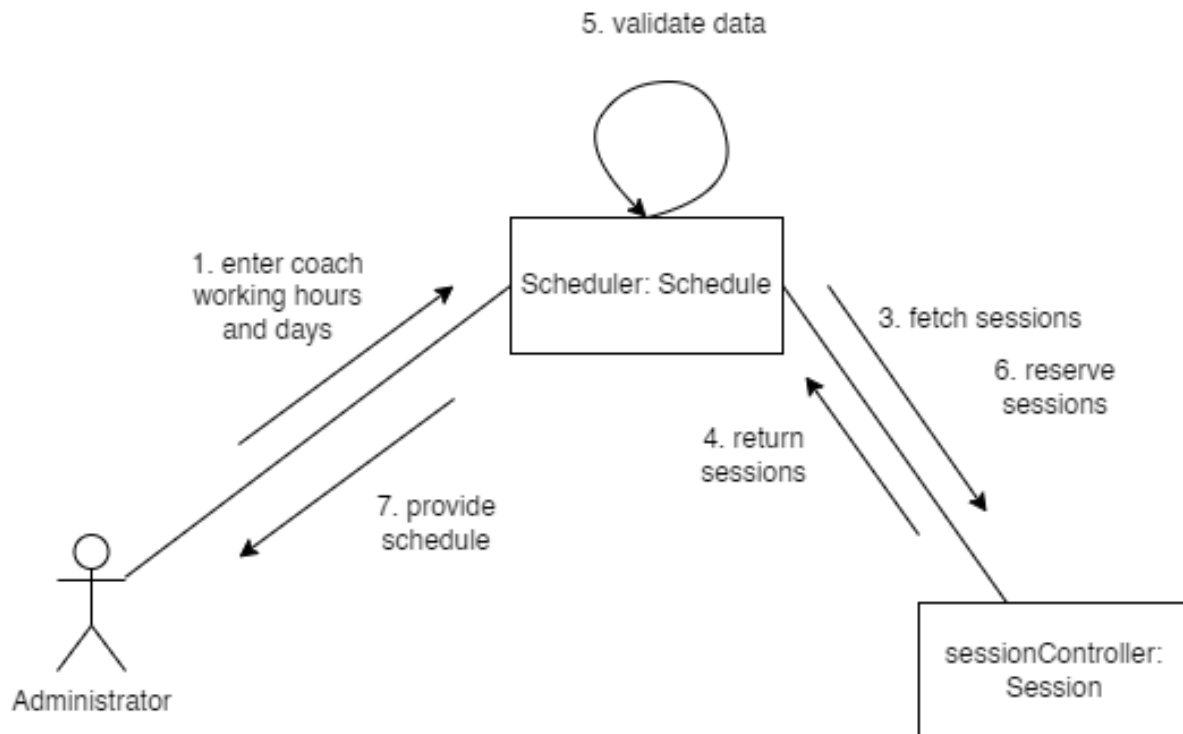
7.0 Class Diagram:

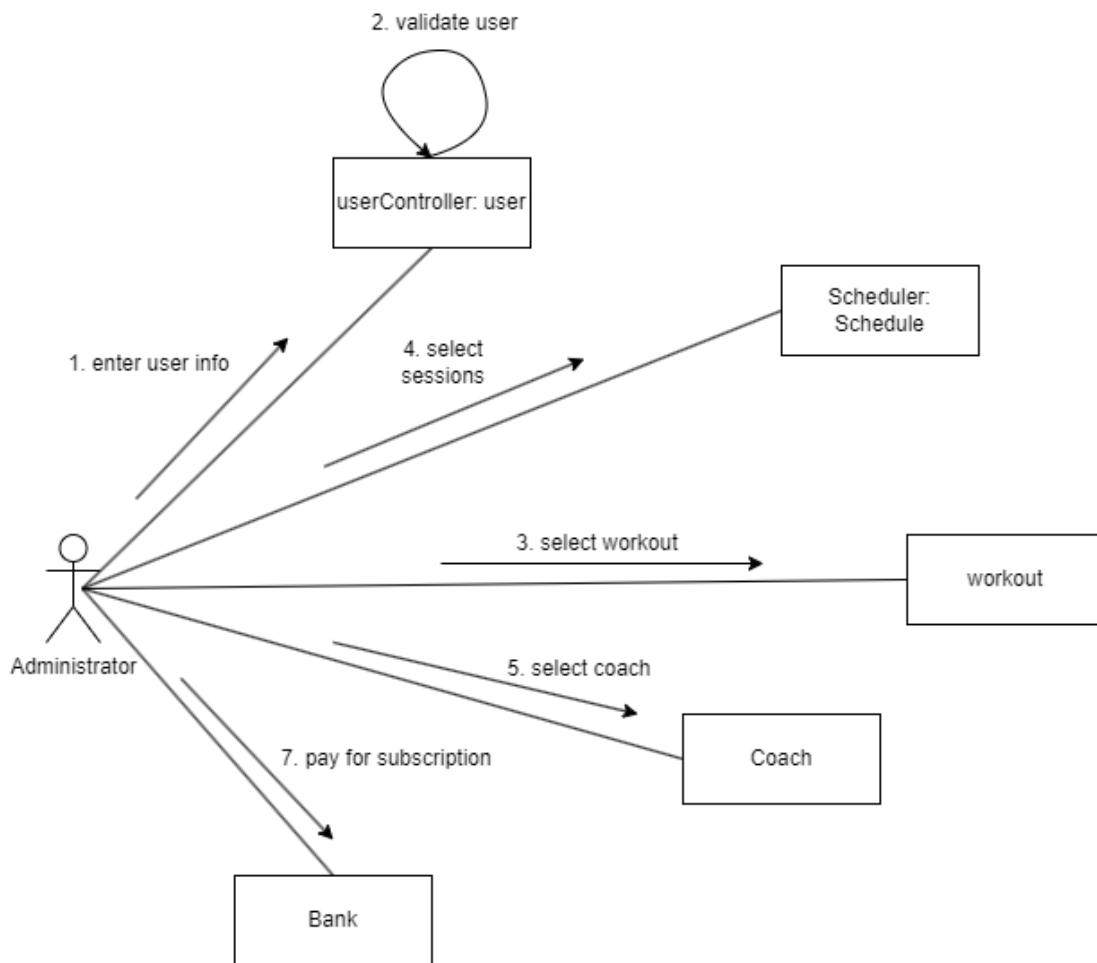


8.0 Collaboration Diagrams:

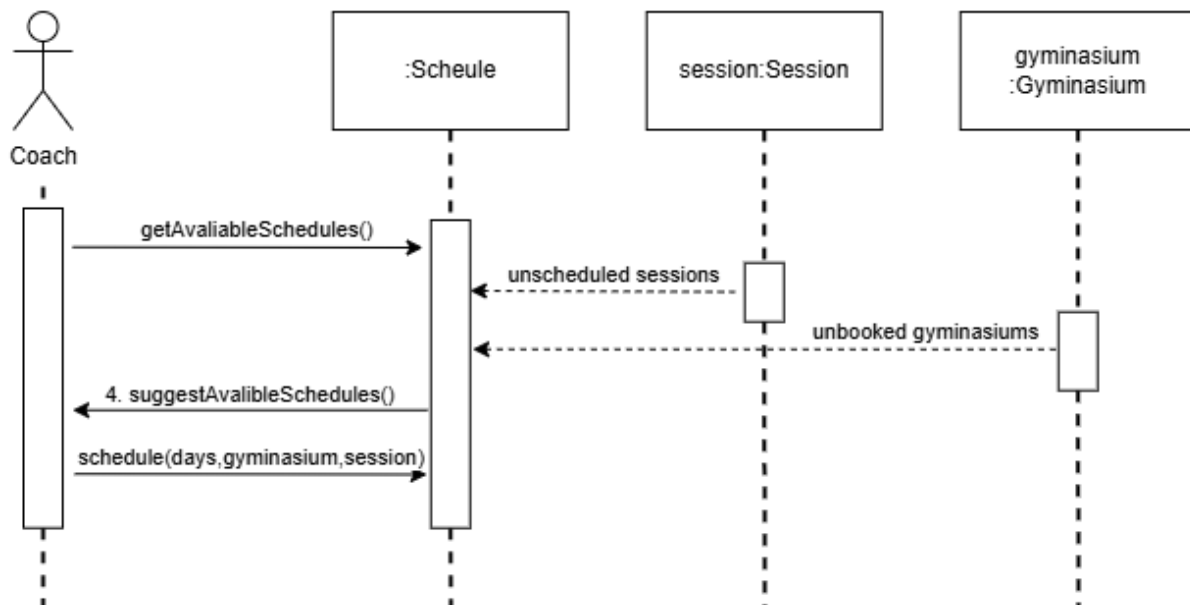
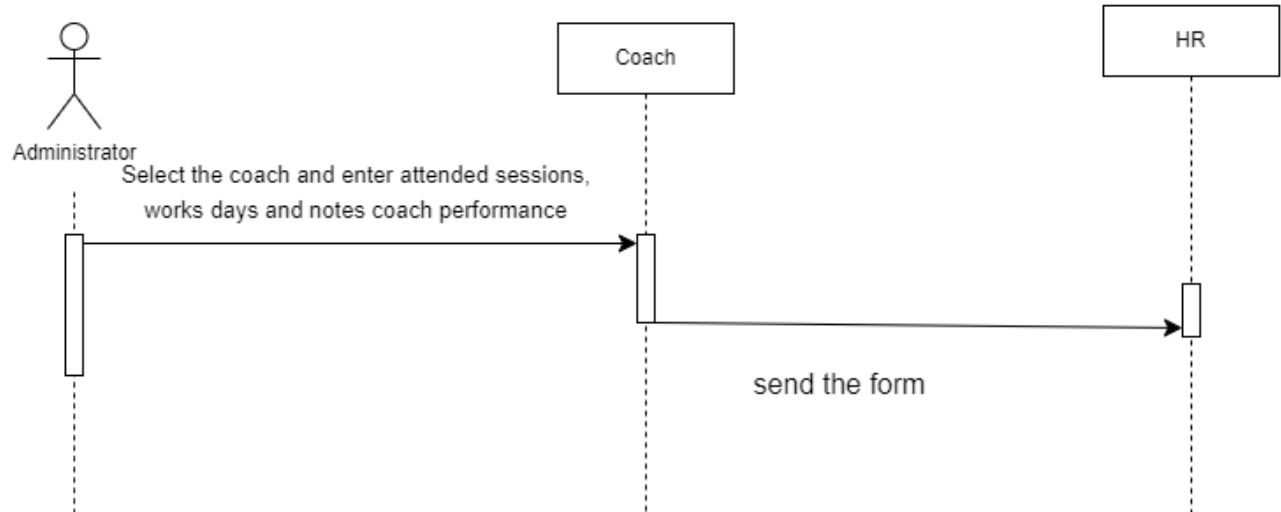


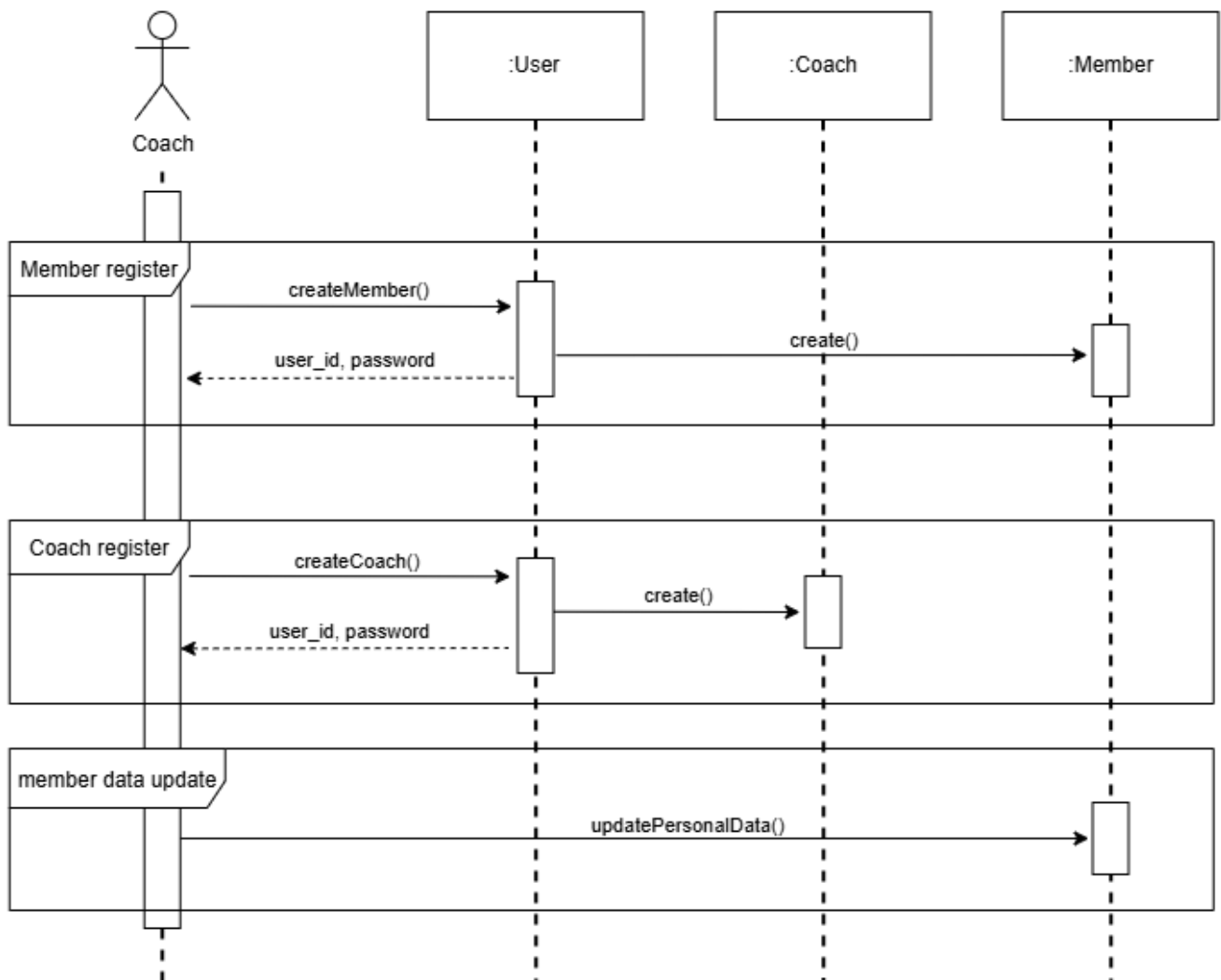


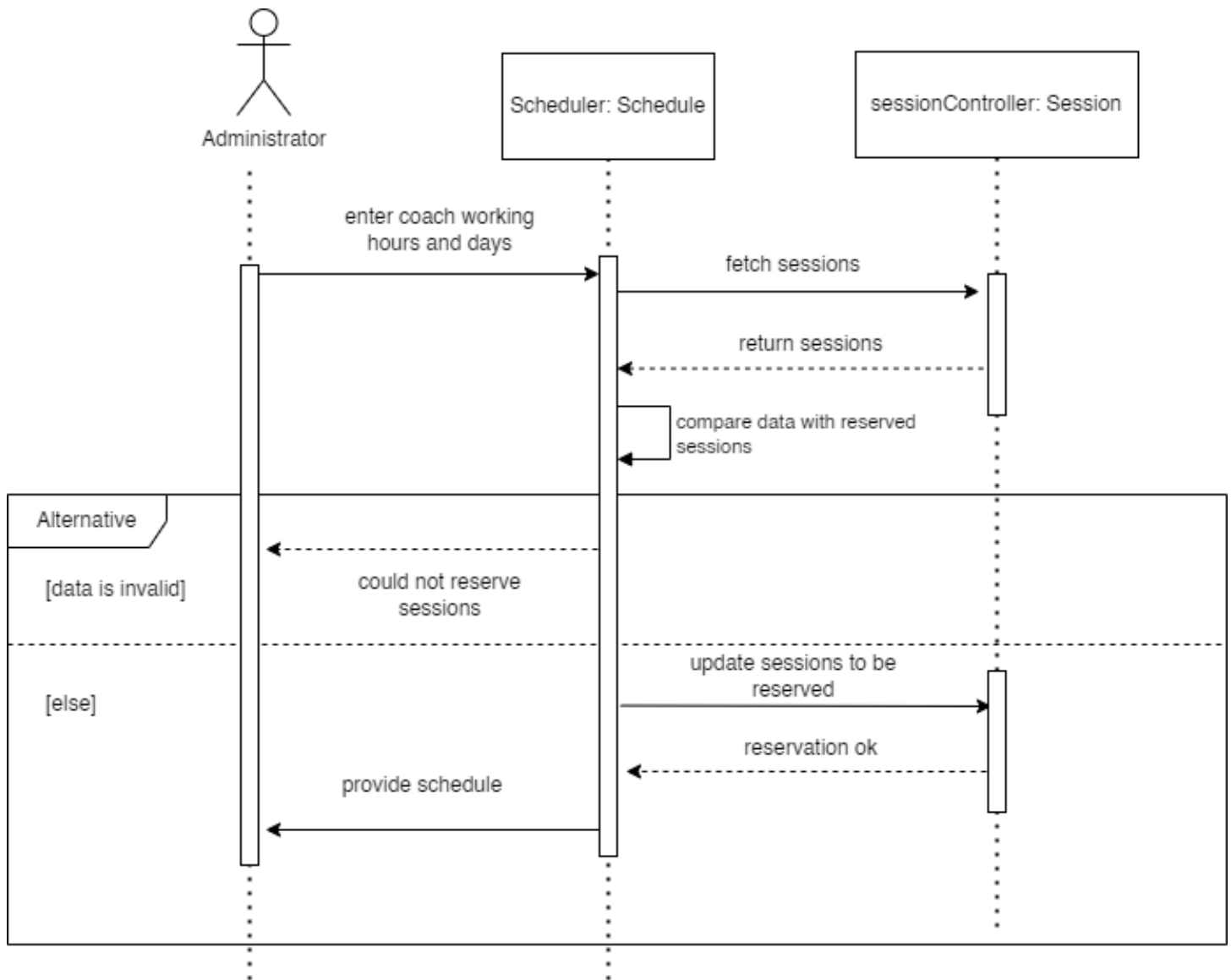


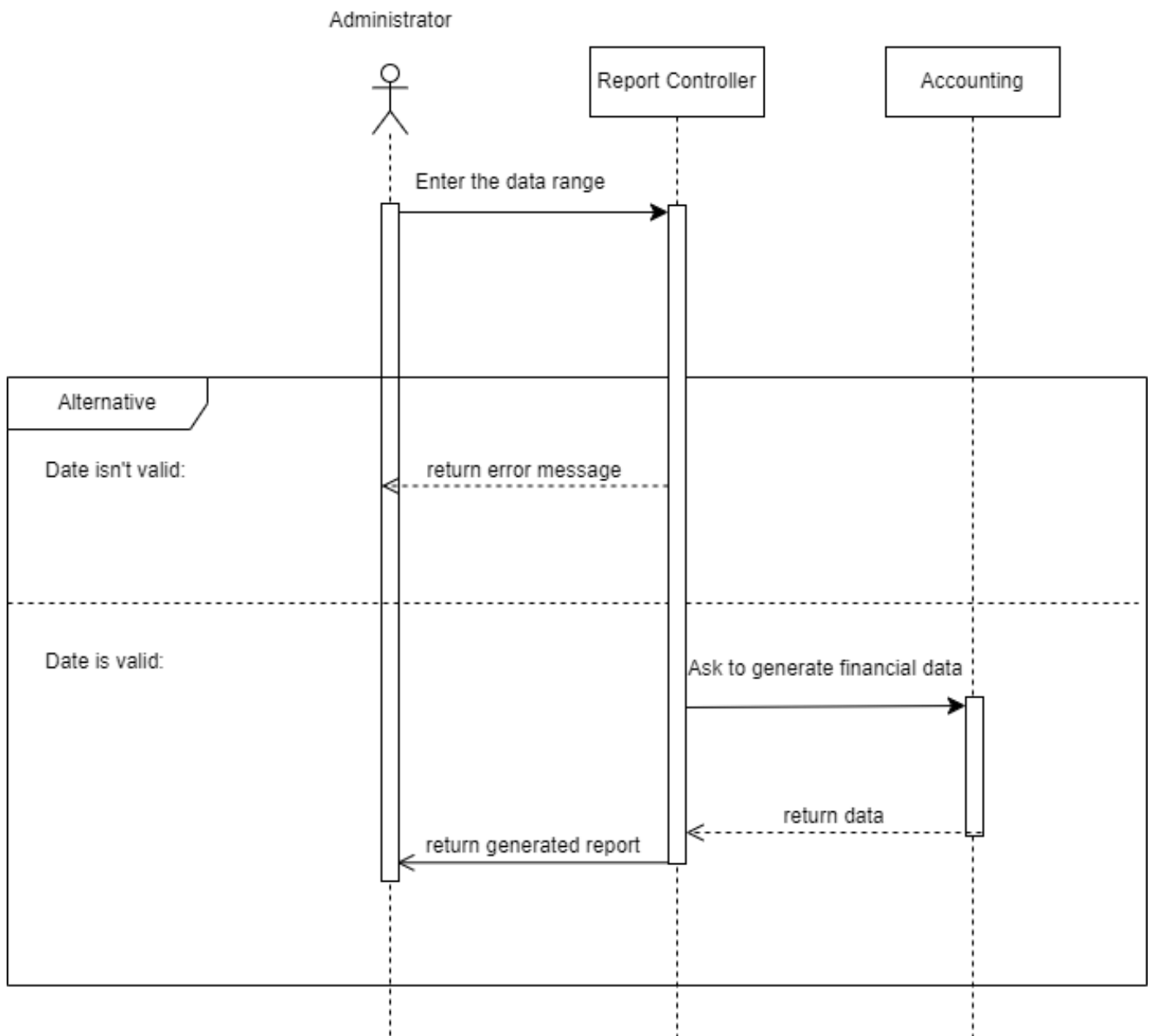


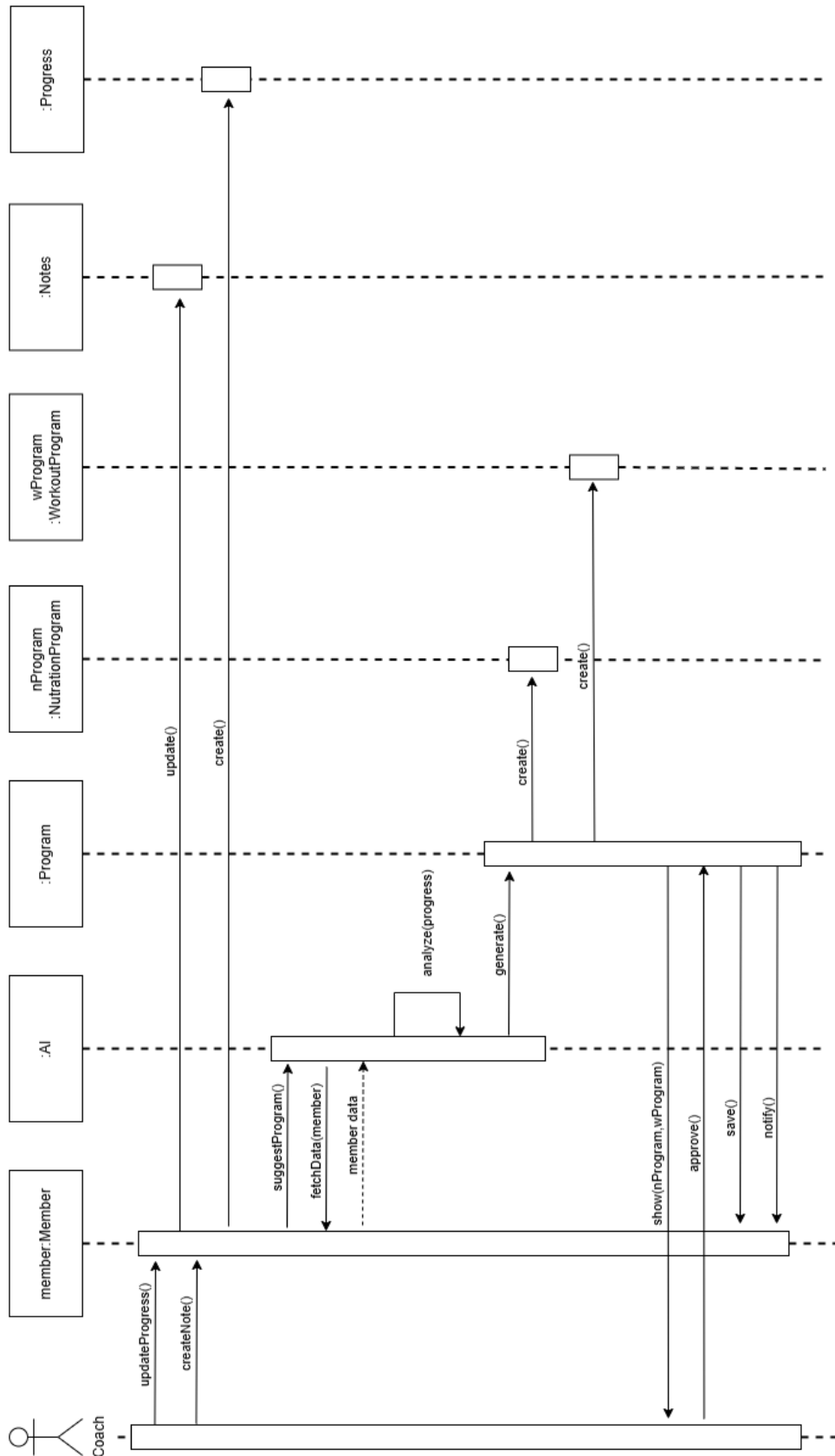
9.0 Sequence Diagrams:

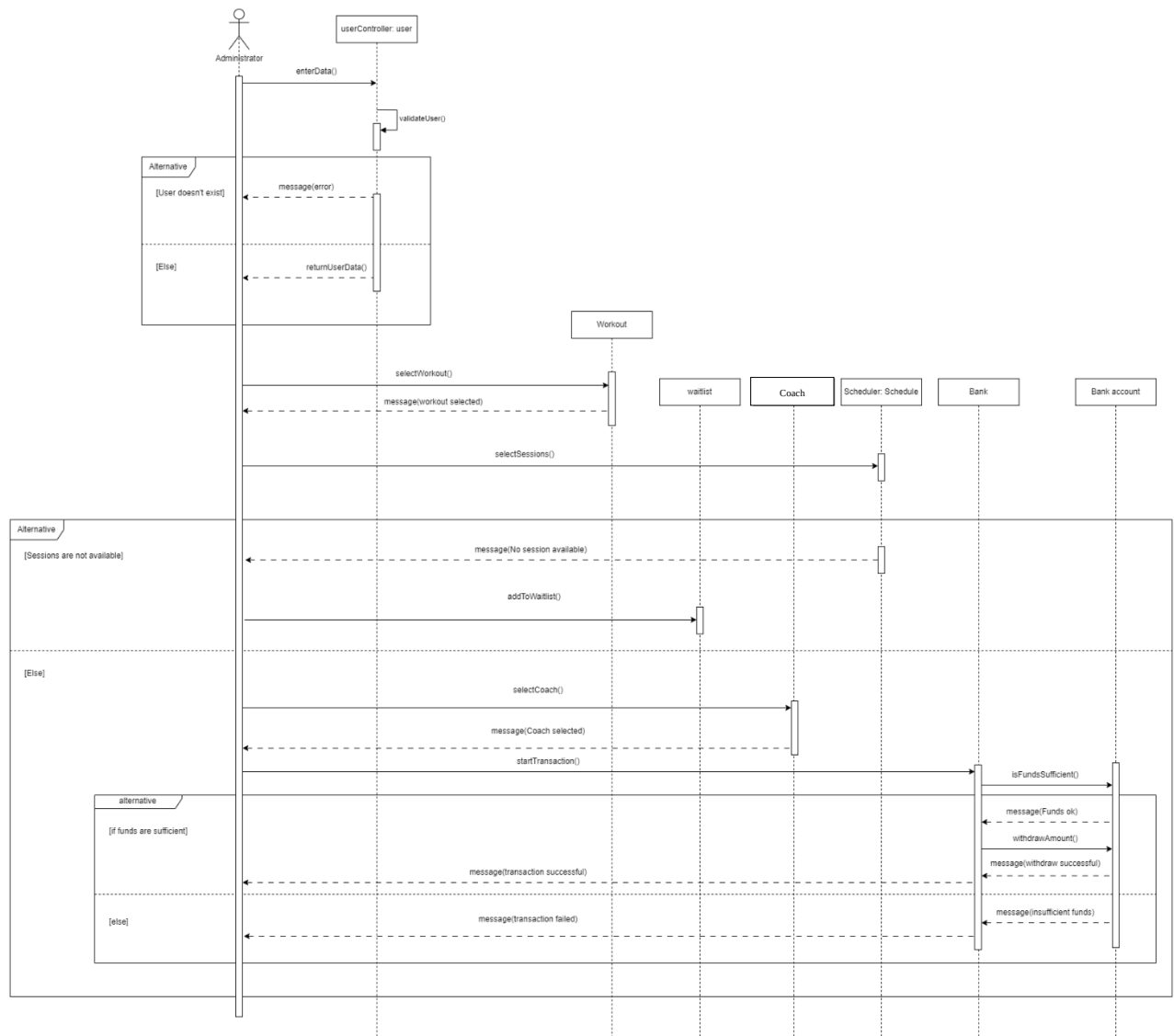




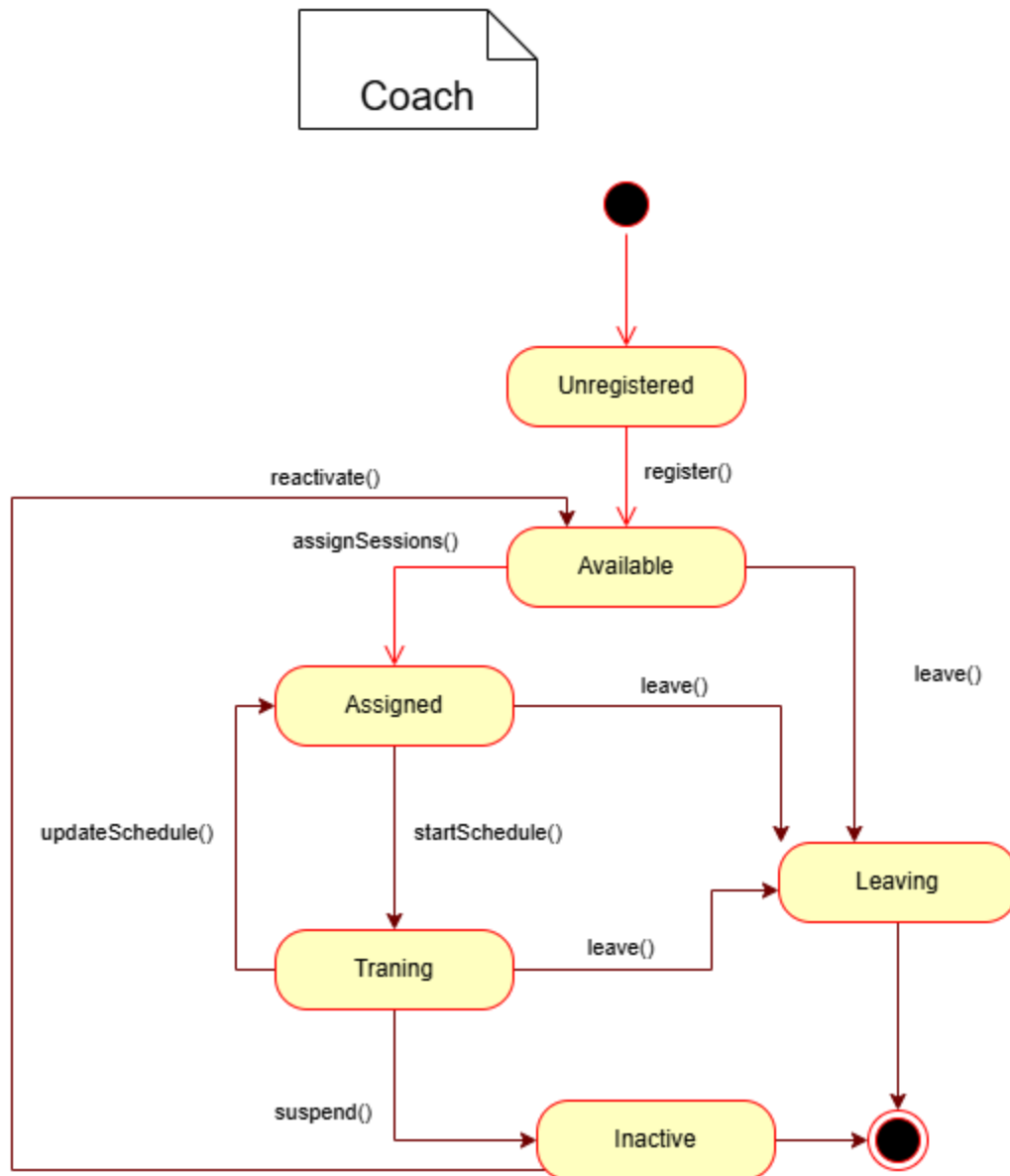








10.0 State Diagrams:



Member Subscription

